LAPORAN PRAKTIKUM PERANCANGAN JARINGAN

Pemilihan dan Penerapan Protokol Switching dan Routing #5

Disusun oleh

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PROGRAM STUDI DIPLOMA IV TEKNOLOGI REKAYASA INTERNET
DEPARTEMEN TEKNIK ELEKTRO DAN INFORMATIKA SEKOLAH VOKASI
UNIVERSITAS GADJAH MADA

2026

TUJUAN

a. Memahami perancangan jaringan secara *top-down*

b. Menerapkan rancangan jaringan berhirarki

c. Melakukan konfigurasi protokol-protokol
   - VLAN
   - Subnetting dan Pengalamatan IP
   - Inter-VLAN Router
   - DHCP
   - SSH
   - RIPv2
   - Switching Security
   - Host

d. Pengujian dan Verifikasi komunikasi jaringan

DASAR TEORI

Faktor-faktor yang dilibatkan dalam sebuah pengambilan keputusan :

- Tujuan harus jelas
- Pilihan-pilihan atau ide yang ada harus dipelajari lebih jauh terlebih dahulu.
- Konsekuensi dari setiap keputusan harus dipelajari lebih lanjut
- Rencana cadangan harus segera dibuat.

Selain faktor-faktor diatas juga harus melibatkan beberapa pertanyaan berikut ini :

- Jika hasil keputusan yang diambil ternyata salah, apa yang akan dikerjakan ?
- Jika hasil keputusan sudah pernah dilakukan sebelumnya, apakah ada permasalahan dengan keputusan tersebut ?
- Bagaimana reaksi pelanggan dengan keputusan anda ?
- Apa rencana cadangan anda jika keputusan anda ditolak oleh pelanggan ?

Berikut ini adalah salah satu contoh dokumen dalam pembuatan keputusan

	Critical Goals			Other Goals		
	Adaptability—must adapt to changes in a large internetwork within seconds	Must scale to a large size (hundreds of routers)	Must be an industry standard and compatible with existing equipment	Should not create a lot of traffic	Should run on inexpensive routers	Should be easy to configure and manage
BGP	X*	X	X	8	7	7
OSPF	X	X	X	8	8	8
IS-IS	X	X	X	8	6	6
IGRP	X	X				
EIGRP	X	X				
RIP			X			

*X = Meets critical criteria. 1 = Lowest. 10 = Highest.

Pemilihan Switching Protocols

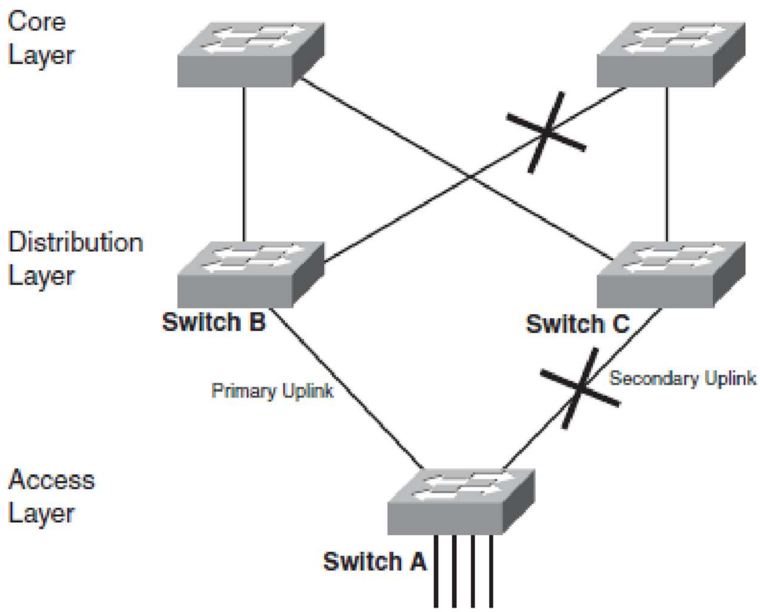
Switch memanfaatkan sirkuit terintegrasi cepat untuk menawarkan latensi rendah. Bridges jauh lebih lambat daripada switch dan memiliki lebih sedikit port dan biaya per port yang lebih tinggi. Untuk alasan ini, switch telah menggantikan bridges. Switch memiliki kemampuan untuk melakukan pemrosesan *store-and-forward* atau pemrosesan *cut-through*. Dengan pemrosesan *cut-through*, sebuah switch dengan cepat melihat alamat tujuan (bidang pertama dalam header frame Ethernet), menentukan port keluar, dan segera mulai mengirim bit ke port keluar. Kerugian dengan pemrosesan *cut-through* adalah meneruskan frame ilegal (misalnya, kerdil Ethernet) dan frame dengan kesalahan CRC.

Pada jaringan yang rawan kegagalan dan kesalahan, pemrosesan *cut-through* tidak boleh digunakan. Beberapa switch memiliki kemampuan untuk secara otomatis berpindah dari mode *cut-through* ke mode *store-and forward* ketika ambang kesalahan tercapai. Fitur ini disebut pengalihan *cut-through adaptif* oleh beberapa vendor. Switch memungkinkan beberapa jalur penerusan paralel, yang berarti sebuah switch dapat menangani volume lalu lintas yang tinggi lebih cepat daripada sebuah bridge. Switch high-end dapat mendukung banyak jalur penerusan simultan, tergantung pada struktur switching.

Berikut ini beberapa protokol switching yang dapat digunakan :

- STP
- RSTP
- PVST
- PVST+
- Etherchannel
- VTP
- DTP

Berikut ini contoh penggunaan STP



Gambar 1 Switch lapisan akses dengan dua uplinks ke switches layer distribusi

Pemilihan Routing Protocols

Protokol routing memungkinkan router secara dinamis mempelajari cara menjangkau jaringan lain dan bertukar informasi ini dengan router atau host lain. Memilih protokol routing untuk pelanggan desain jaringan Anda lebih sulit daripada memilih protokol switching, karena ada begitu banyak pilihan. Keputusan akan lebih mudah jika Anda dapat menggunakan tabel keputusan, seperti yang ditunjukkan pada Tabel 7-1 pada buku *Top-Down Network Design 3rd Edition.* Berbekal pemahaman yang kuat tentang tujuan dan informasi pelanggan Anda tentang karakteristik protokol routing yang berbeda, Anda dapat membuat keputusan yang tepat tentang protokol routing mana yang direkomendasikan. Berikut ini tabel ringkasan panduan dalam pemilihan protocol routing

Tabel 1 Perbandingan protokol-protokol routing

	Distance Vector or Link State	Interior or Exterior	Classful or Classless	Metrics Supported	Scalability	Convergence Time	Resource Consumption	Supports Security? Authenticates Routes?	Ease of Design, Configuration, and Troubleshooting
RIPv1	Distance vector	Interior	Classful	Hop count	15 hops	Can be long (if no load balancing)	Memory: low CPU: low Bandwidth: high	No	Easy
RIPv2	Distance vector	Interior	Classless	Hop count	15 hops	Can be long (if no load balancing)	Memory: low CPU: low Bandwidth: high	Yes	Easy
IGRP	Distance vector	Interior	Classful	Bandwidth, delay, reliability, load	255 hops (default is 100)	Quick (uses triggered updates and poison reverse)	Memory: low CPU: low Bandwidth: high	No	Easy
EIGRP	Advanced distance vector	Interior	Classless	Bandwidth, delay, reliability, load	1000s of routers	Very quick (uses DUAL algorithm)	Memory: moderate CPU: low Bandwidth: low	Yes	Easy
OSPF	Link state	Interior	Classless	Cost (100 million divided by bandwidth on Cisco routers)	A few hundred routers per area, a few hundred areas	Quick (uses LSAs and Hello packets)	Memory: high CPU: high Bandwidth: low	Yes	Moderate
BGP	Path vector	Exterior	Classless	Value of path attributes and other configurable factors	1000s of routers	Quick (uses update and keepalive packets, and withdraws routes)	Memory: high CPU: high Bandwidth: low	Yes	Moderate
IS-IS	Link state	Interior	Classless	Configured path value, plus delay, expense, and errors	Hundreds of routers per area, a few hundred areas	Quick (uses LSAs)	Memory: high CPU: high Bandwidth: low	Yes	Moderate

Berikut juga adalah tabel administrative distance berdasarkan tipe route untuk preferensi penentuan rute oleh router.

Tabel 2 Administrative Distance berdasarkan tipe route

Route Source	Default Distance Value
Connected interface	0
Static route	1
Enhanced Interior Gateway Routing Protocol (EIGRP) summary route	5
External Border Gateway Protocol (BGP)	20
Internal EIGRP	90
IGRP	100
OSPF	110
Intermediate System-to-Intermediate System (IS-IS)	115
Routing Information Protocol (RIP)	120
Exterior Gateway Protocol (EGP)	140
On-Demand Routing (ODR)	160
External EIGRP	170
Internal BGP	200
Unknown	255

Simpulan Pemilihan Protokol

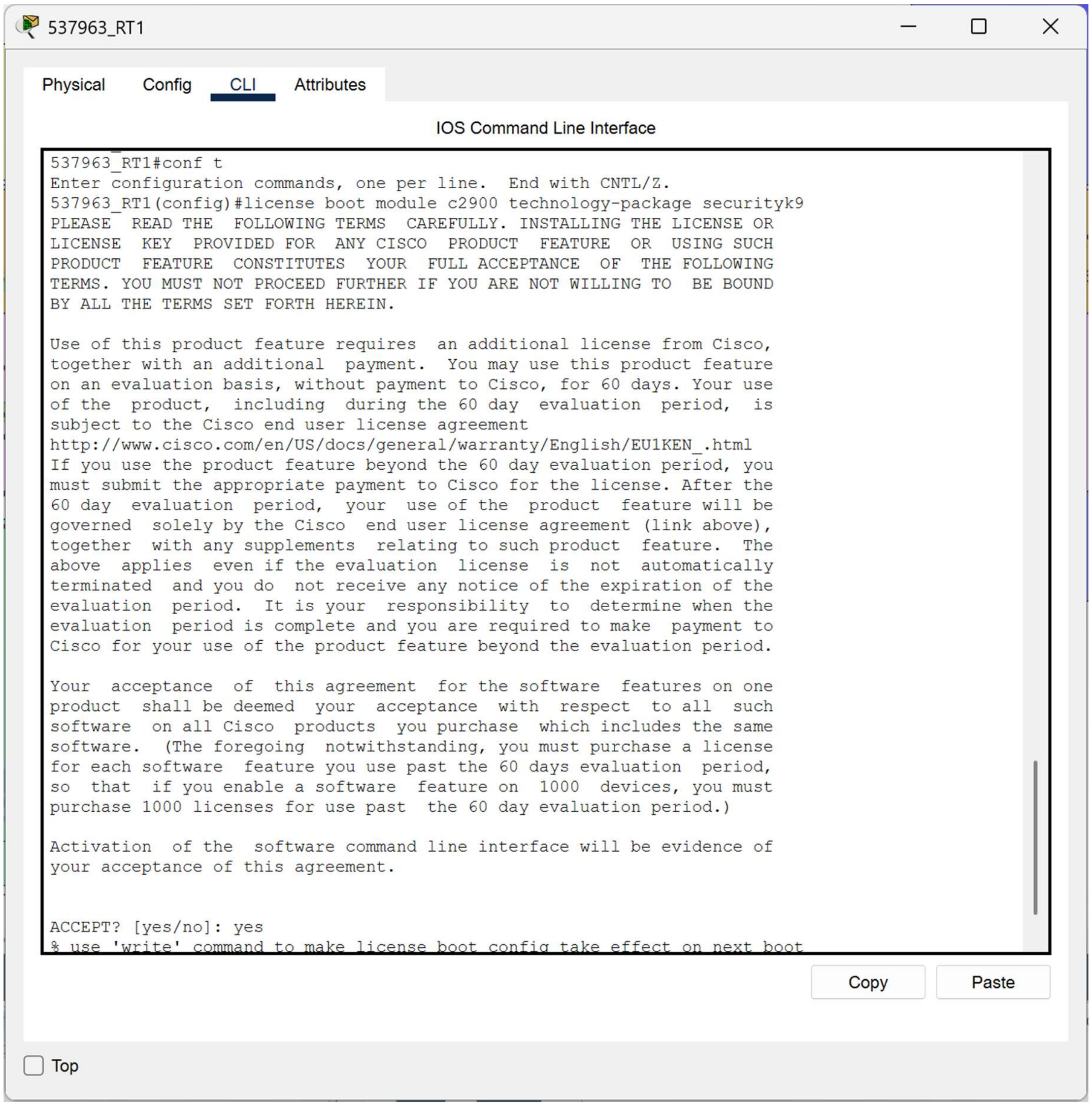
Pemilihan protokol switching dan routing yang tepat untuk pelanggan desain jaringan Anda sangat diperlukan. Skalabilitas dan karakteristik kinerja protokol dan seberapa cepat protokol dapat beradaptasi dengan perubahan menjadi perhatian penting dalam pemilihan protocol witching maupun routing.

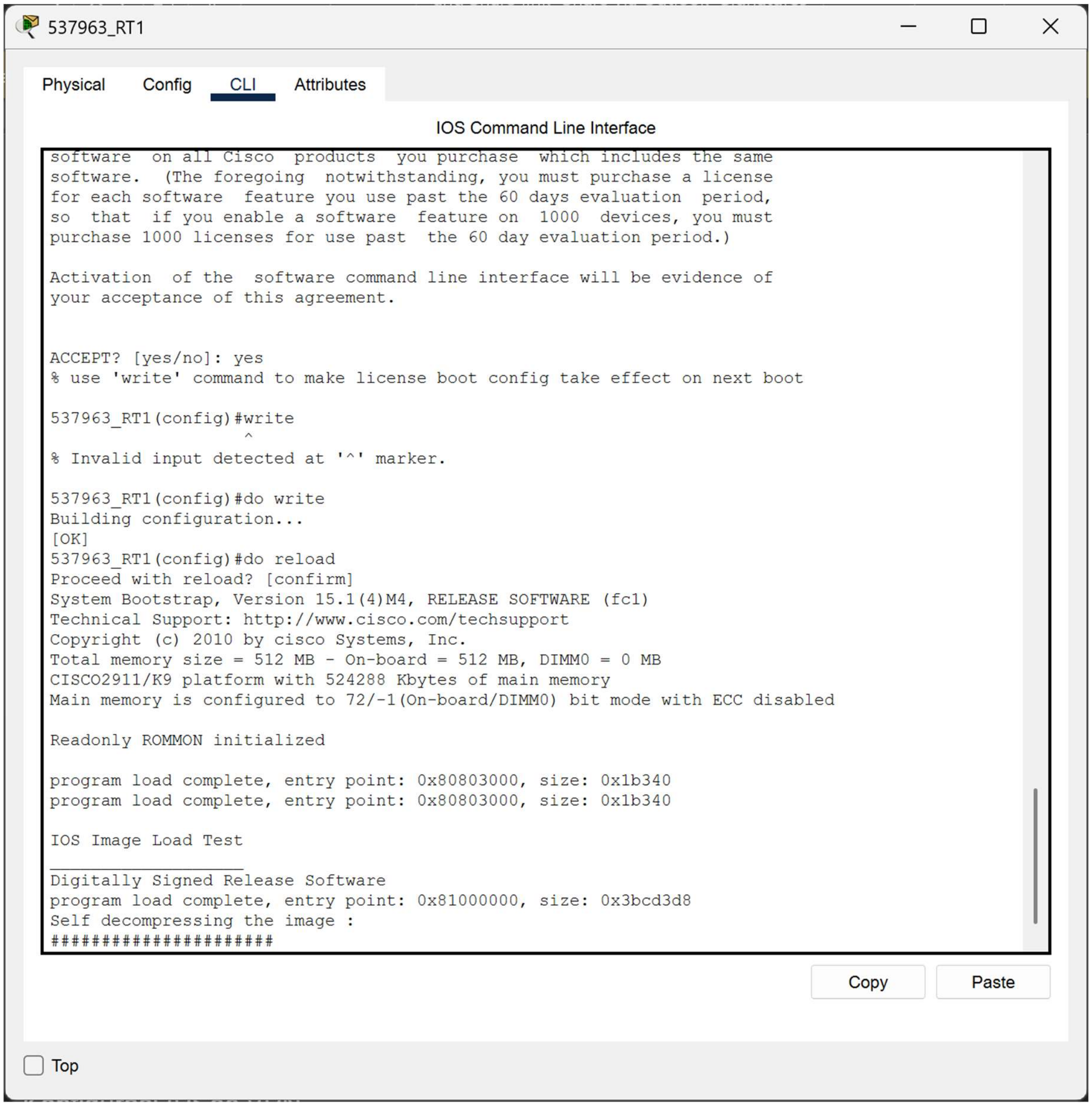
Memutuskan protokol switching dan routing yang tepat untuk pelanggan Anda akan membantu Anda memilih produk switch dan router terbaik untuk pelanggan. Misalnya, jika Anda telah memutuskan bahwa desain harus mendukung protokol routing yang dapat menyatu dalam hitungan detik di internetwork besar, Anda mungkin tidak akan merekomendasikan router yang hanya menjalankan RIP.

Detail pembahasan bagaimana memilih protocol switching dan routing sangaat direkomendasikan untuk membaca buku Top-down Network Design 3rd Edition dari CISCO yang juga sebagai buku utama matakuliah teori Perancangan Jaringan. Buku tersebut terdapat diskusi umum tentang pengambilan keputusan untuk membantu Anda mengembangkan proses sistematis untuk memilih solusi desain jaringan. Sebuah diskusi tentang menjembatani dan beralih protokol diikuti, yang meliputi menjembatani transparan, beralih multilayer, perangkat tambahan untuk protokol STP, dan VLAN. Bagian pada protokol routing mengikuti bagian switching.

PROSEDUR PERCOBAAN

1. Aktivasi Lisensi Security




537963_RT1
Physical
Config
CLI
Attributes
IOS Command Line Interface
software on all Cisco products you purchase which includes the same
software. (The foregoing notwithstanding, you must purchase a license
for each software feature you use past the 60 days evaluation period,
so that if you enable a software feature on 1000 devices, you must
purchase 1000 licenses for use past the 60 day evaluation period.)

Activation of the software command line interface will be evidence of
your acceptance of this agreement.

ACCEPT? [yes/no]: yes
% use 'write' command to make license boot config take effect on next boot

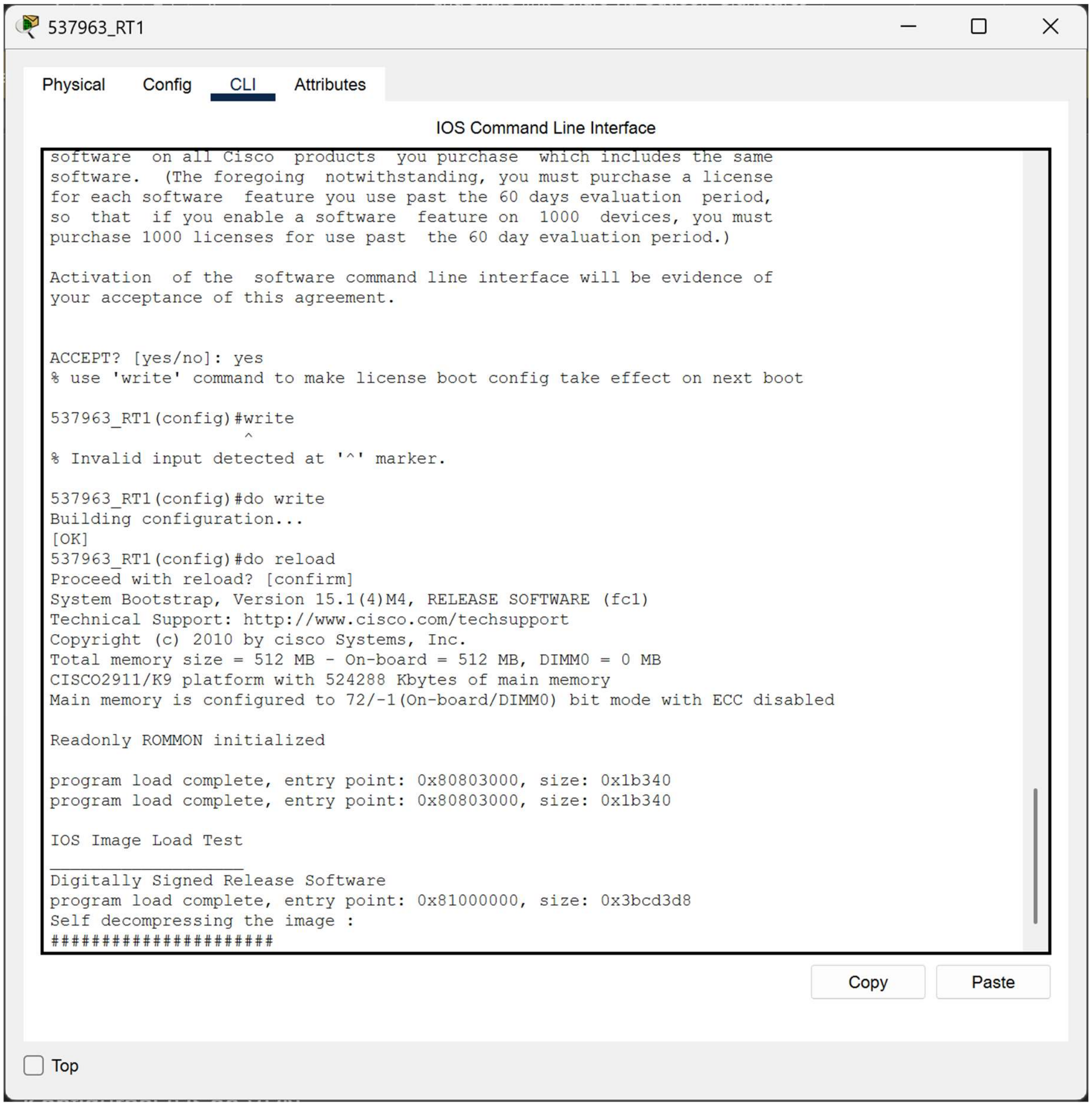
537963_RT1(config)#write
^
% Invalid input detected at '^' marker.

537963_RT1(config)#do write
Building configuration...
[OK]
537963_RT1(config)#do reload
Proceed with reload? [confirm]
System Bootstrap, Version 15.1(4)M4, RELEASE SOFTWARE (fc1)
Technical Support: http://www.cisco.com/techsupport
Copyright (c) 2010 by cisco Systems, Inc.
Total memory size = 512 MB - On-board = 512 MB, DIMM0 = 0 MB
CISCO2911/K9 platform with 524288 Kbytes of main memory
Main memory is configured to 72/-1(On-board/DIMM0) bit mode with ECC disabled

Readonly ROMMON initialized

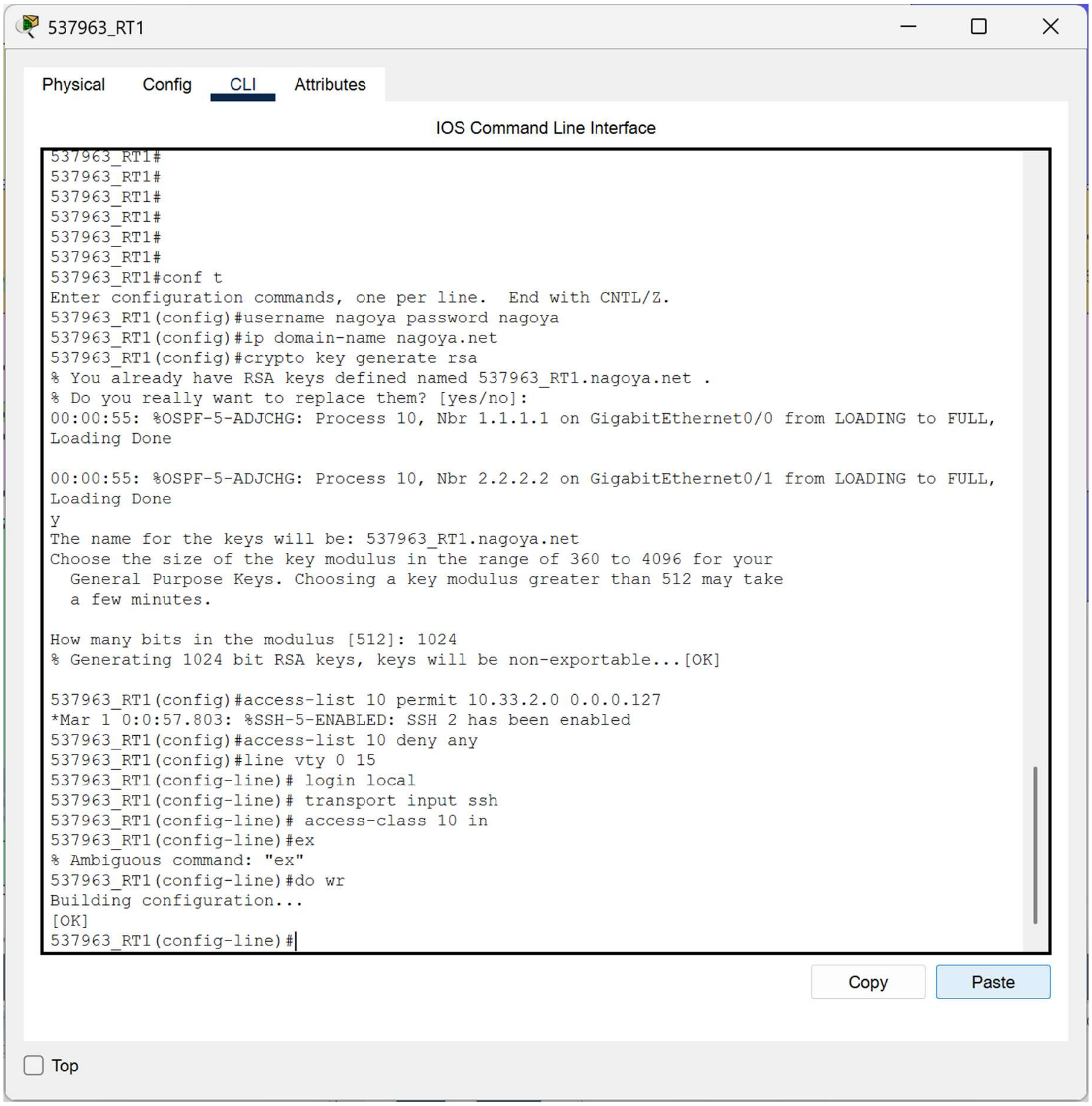
program load complete, entry point: 0x80803000, size: 0x1b340
program load complete, entry point: 0x80803000, size: 0x1b340

IOS Image Load Test
____
Digitally Signed Release Software
program load complete, entry point: 0x81000000, size: 0x3bcd3d8
Self decompressing the image :
######################
Copy
Paste
Top



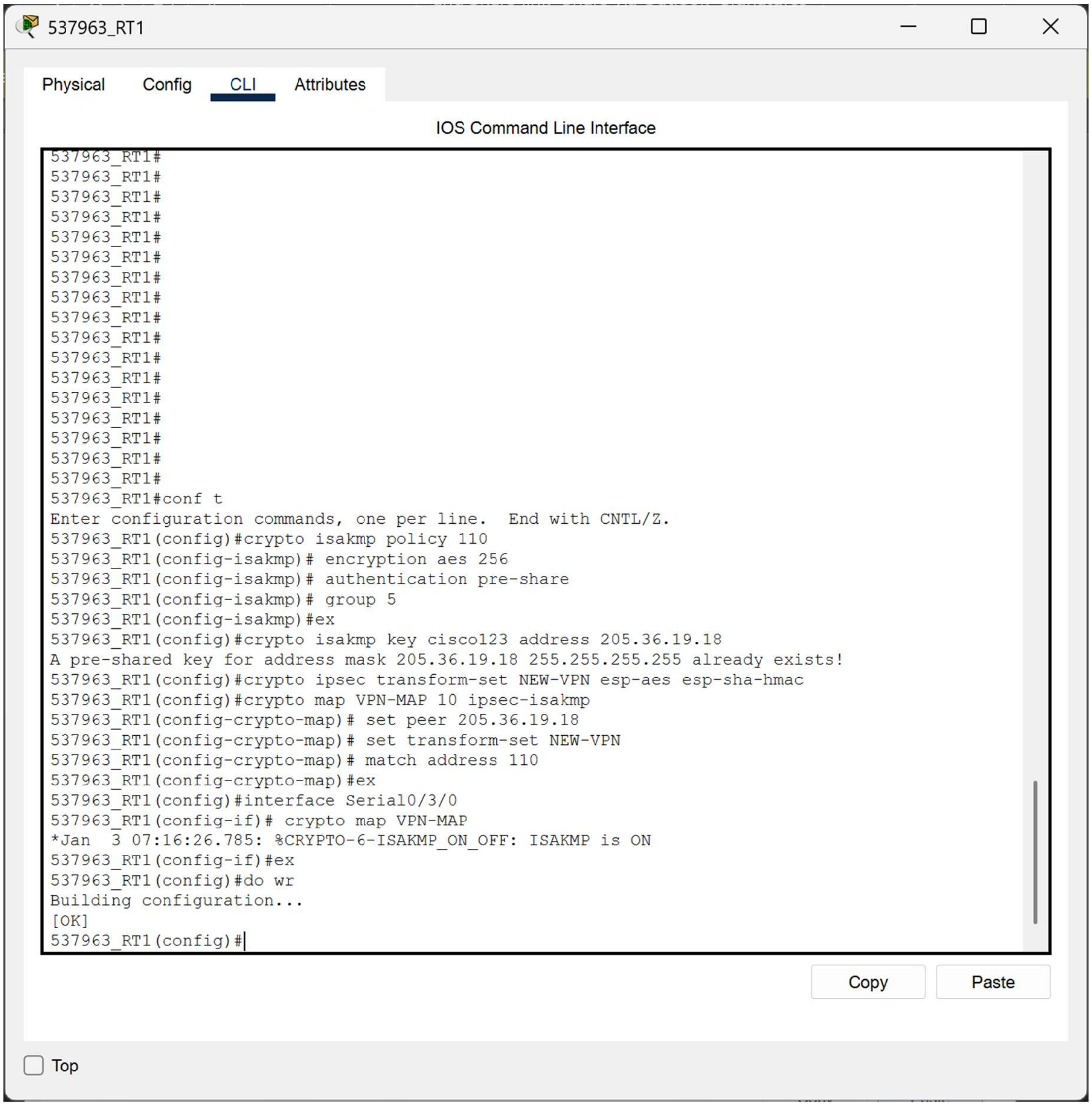
Tahap ini bertujuan untuk mengaktifkan fitur keamanan pada perangkat Cisco seri 2900. Secara default, fitur enkripsi untuk layanan seperti IPSec VPN dan SSH ter-disable. Dengan menjalankan perintah license boot dan menyetujui End User License Agreement (EULA), router akan membuka kapabilitas keamanan tersebut. Proses write dan reload dilakukan agar sistem operasi router (IOS) memuat ulang paket teknologi yang baru diaktifkan tersebut dari memory.

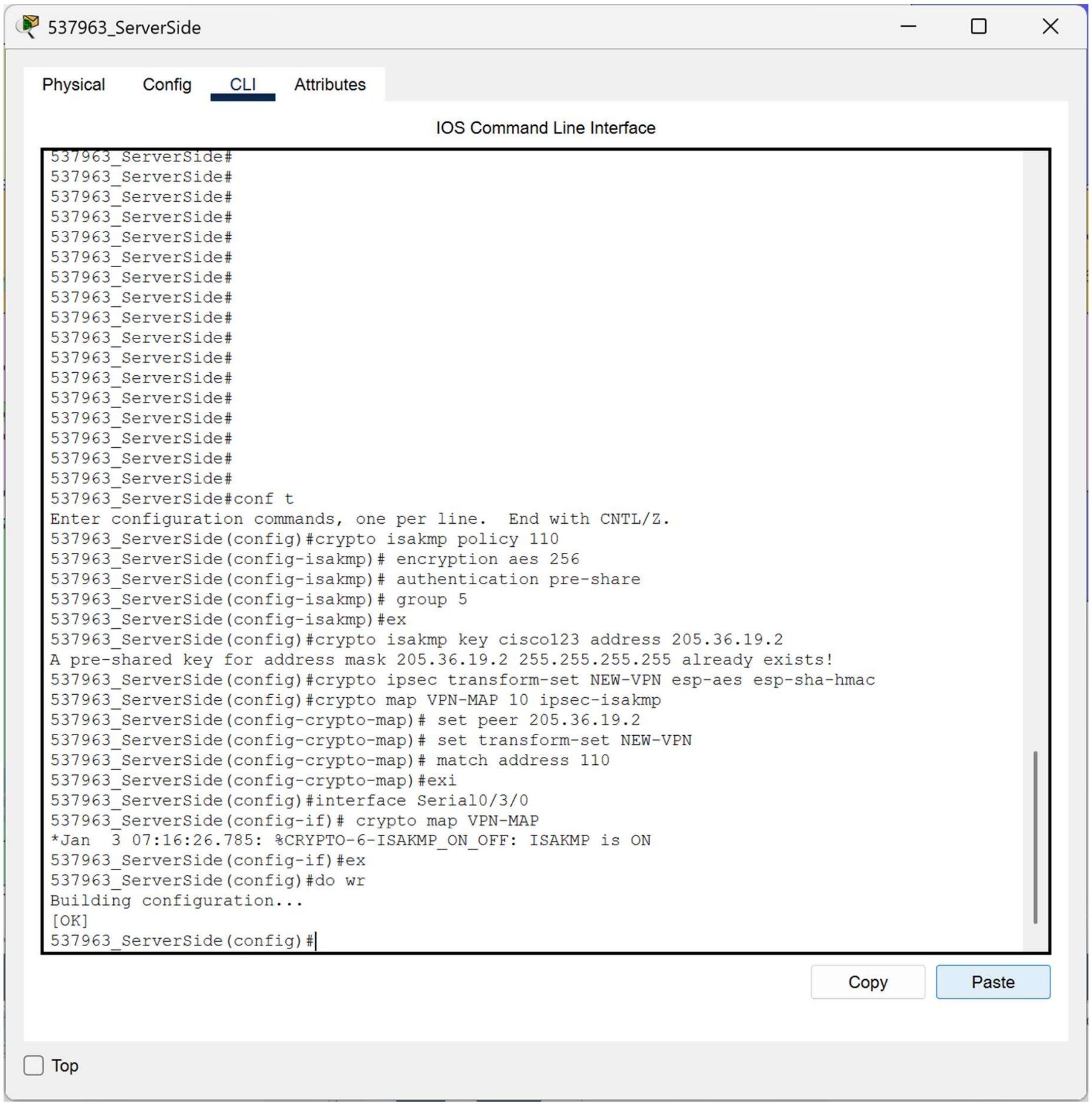
2. Konfigurasi SSH dan ACL



Langkah ini merupakan konfigurasi SSH dan ACL. Ini digunakan untuk mengamankan jalur remote management. Penggunaan SSH menjamin enkripsi data saat login, sementara access-class 10 memastikan hanya perangkat dengan IP yang terdaftar di ACL 10 yang diizinkan mengakses router.

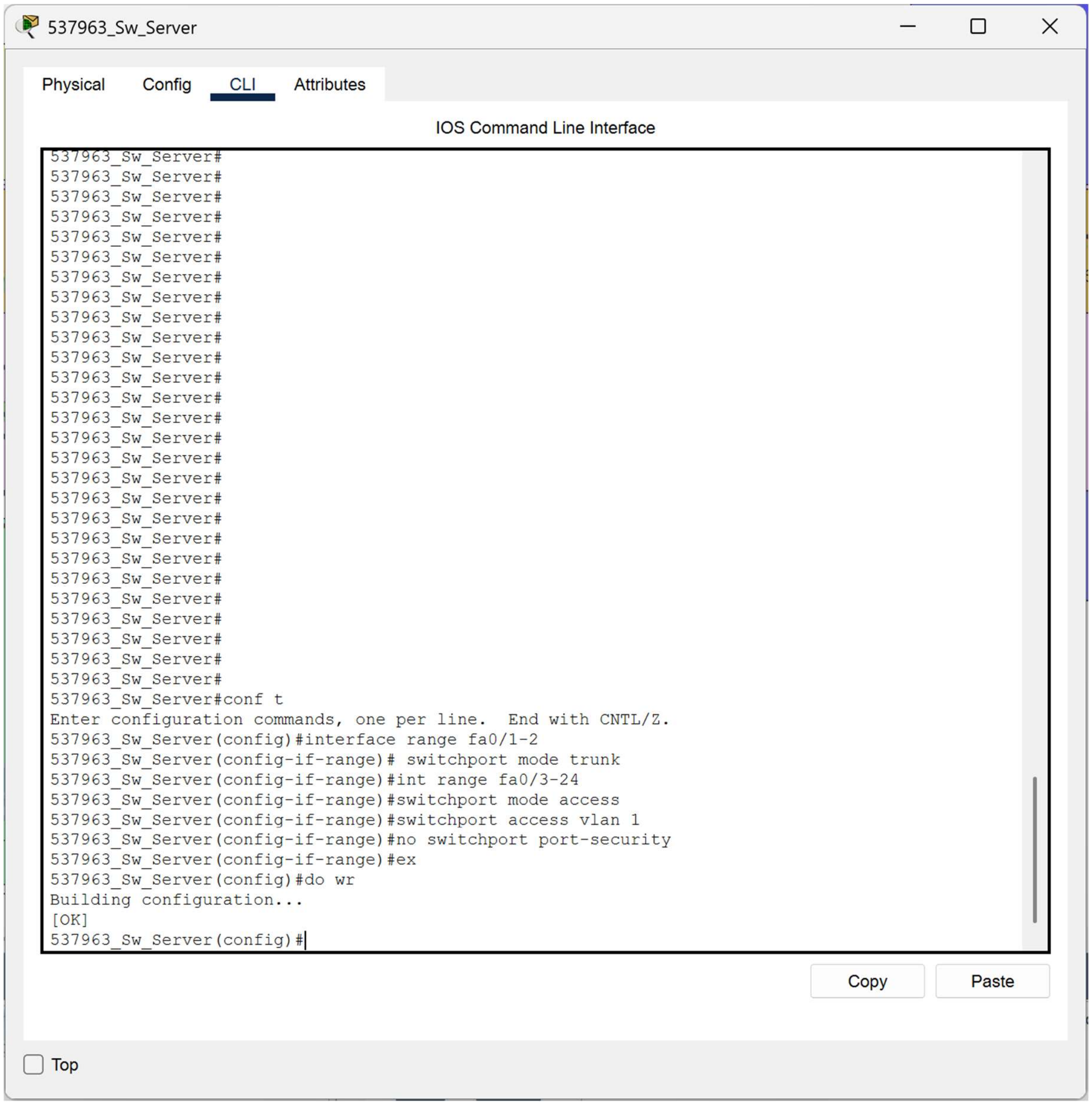
3. Konfigurasi IPSec VPN


537963_RT1
Physical
Config
CLI
Attributes
IOS Command Line Interface
537963_RT1#
537963_RT1#
537963_RT1#
537963_RT1#
537963_RT1#
537963_RT1#
537963_RT1#
537963_RT1#
537963_RT1#
537963_RT1#
537963_RT1#
537963_RT1#
537963_RT1#
537963_RT1#
537963_RT1#
537963_RT1#
537963_RT1#
537963_RT1#conf t
Enter configuration commands, one per line. End with CNTL/Z.
537963_RT1(config)#crypto isakmp policy 110
537963_RT1(config-isakmp)# encryption aes 256
537963_RT1(config-isakmp)# authentication pre-share
537963_RT1(config-isakmp)# group 5
537963_RT1(config-isakmp)#ex
537963_RT1(config)#crypto isakmp key cisco123 address 205.36.19.18
A pre-shared key for address mask 205.36.19.18 255.255.255.255 already exists!
537963_RT1(config)#crypto ipsec transform-set NEW-VPN esp-aes esp-sha-hmac
537963_RT1(config)#crypto map VPN-MAP 10 ipsec-isakmp
537963_RT1(config-crypto-map)# set peer 205.36.19.18
537963_RT1(config-crypto-map)# set transform-set NEW-VPN
537963_RT1(config-crypto-map)# match address 110
537963_RT1(config-crypto-map)#ex
537963_RT1(config)#interface Serial0/3/0
537963_RT1(config-if)# crypto map VPN-MAP
*Jan 3 07:16:26.785: %CRYPTO-6-ISAKMP_ON_OFF: ISAKMP is ON
537963_RT1(config-if)#ex
537963_RT1(config)#do wr
Building configuration...
[OK]
537963_RT1(config)#
Copy
Paste
Top



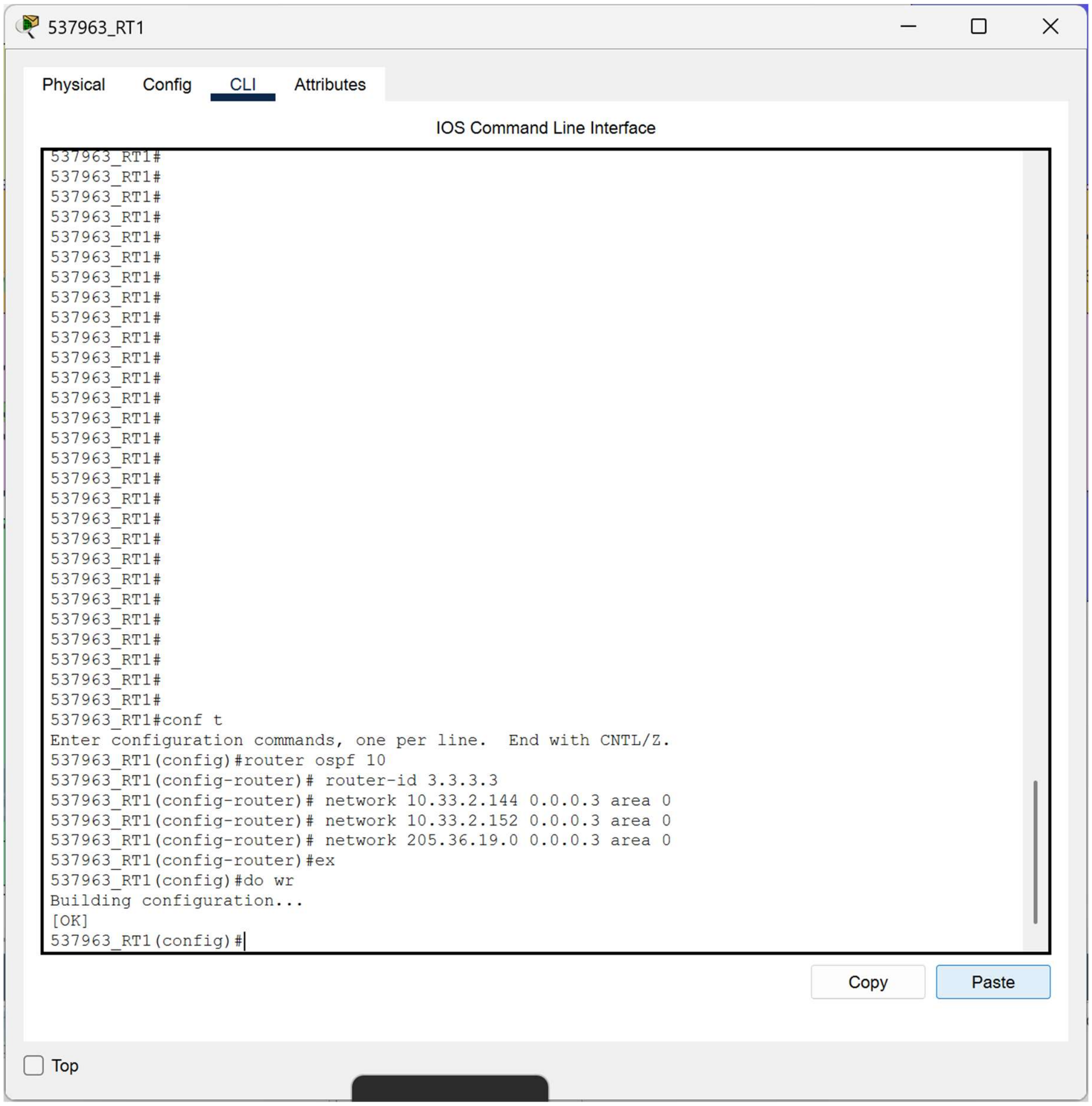
Konfigurasi ini membangun tunnel aman di atas jaringan publik. Terdapat dua fase. Fase pertama menentukan metode autentikasi, sedangkan fase kedua dan crypto map menentukan data mana yang harus dienkripsi (berdasarkan ACL 110) sebelum dikirim ke IP tujuan.

4. Konfigurasi VLAN dan Trunk



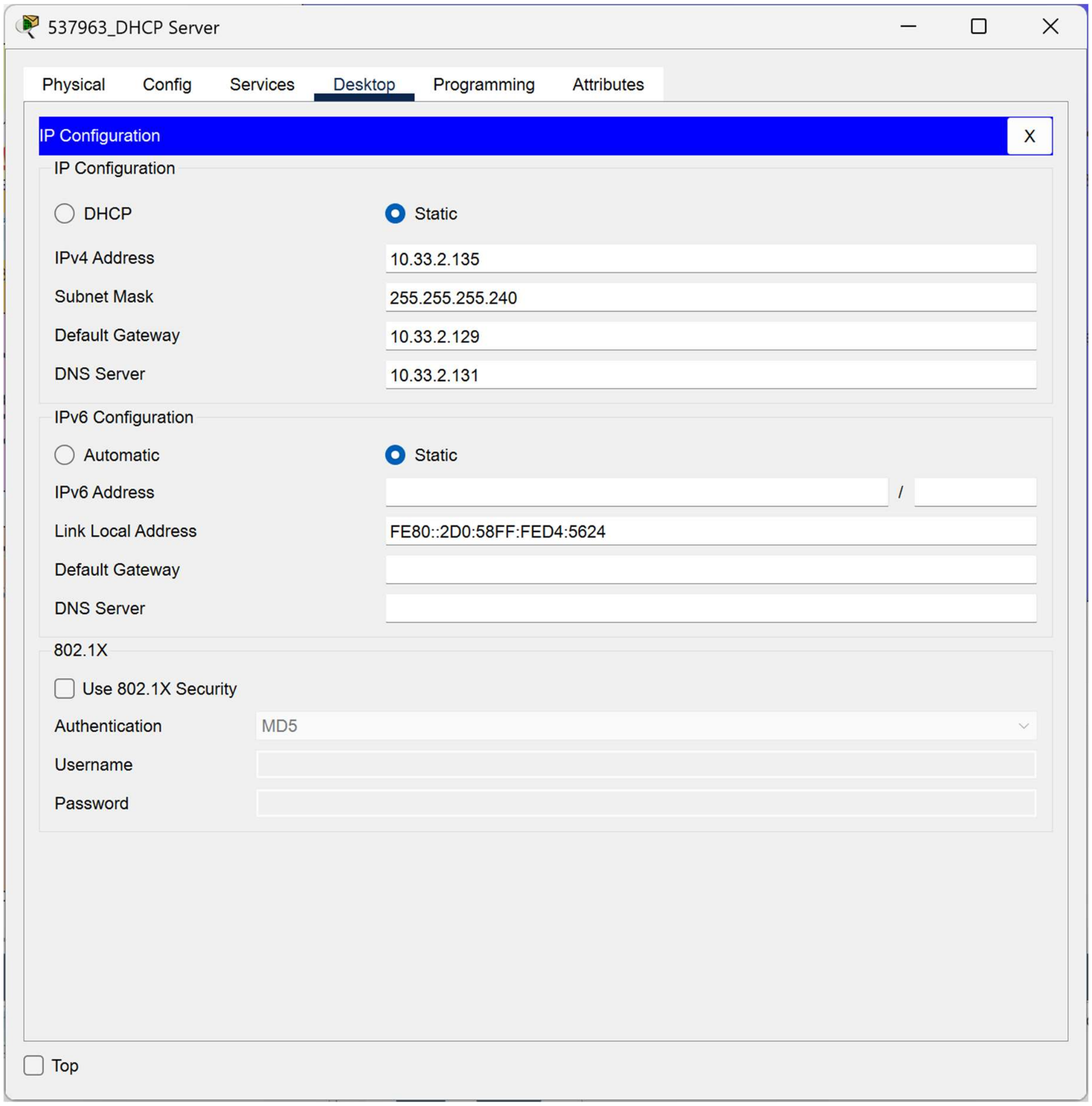
Port trunk dikonfigurasi pada jalur uplink untuk memungkinkan pengiriman paket dengan tag VLAN yang berbeda. Sementara port access digunakan untuk perangkat akhir (seperti DHCP Server) agar dapat berkomunikasi langsung di dalam jaringan lokal.

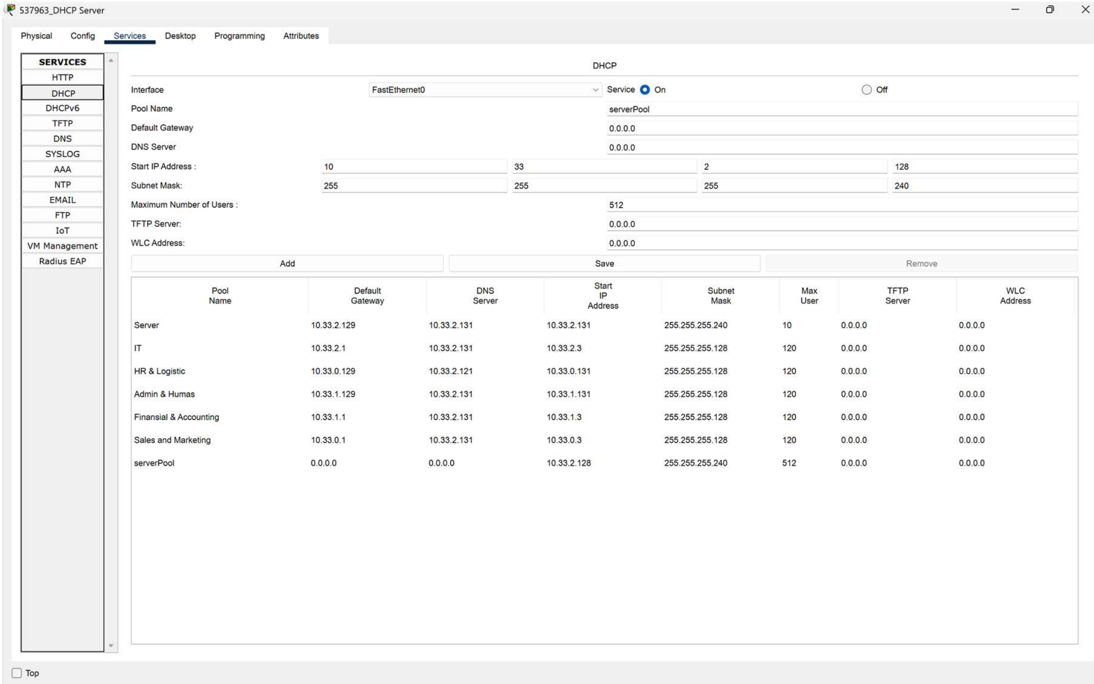
5. Konfigurasi Routing OSPF



OSPF digunakan sebagai protokol routing internal untuk mempercepat konvergensi jaringan. Setiap router akan saling bertukar informasi tabel routing sehingga paket dapat menemukan jalur terbaik menuju tujuannya secara dinamis.

6. Konfigurasi DHCP Server


537963_DHCP Server
Physical
Config
Services
Desktop
Programming
Attributes
IP Configuration
X
IP Configuration
DHCP
Static
IPv4 Address
10.33.2.135
Subnet Mask
255.255.255.240
Default Gateway
10.33.2.129
DNS Server
10.33.2.131
IPv6 Configuration
Automatic
Static
IPv6 Address
/
Link Local Address
FE80::2D0:58FF:FED4:5624
Default Gateway
DNS Server
802.1X
Use 802.1X Security
Authentication
MD5
Username
Password
Top

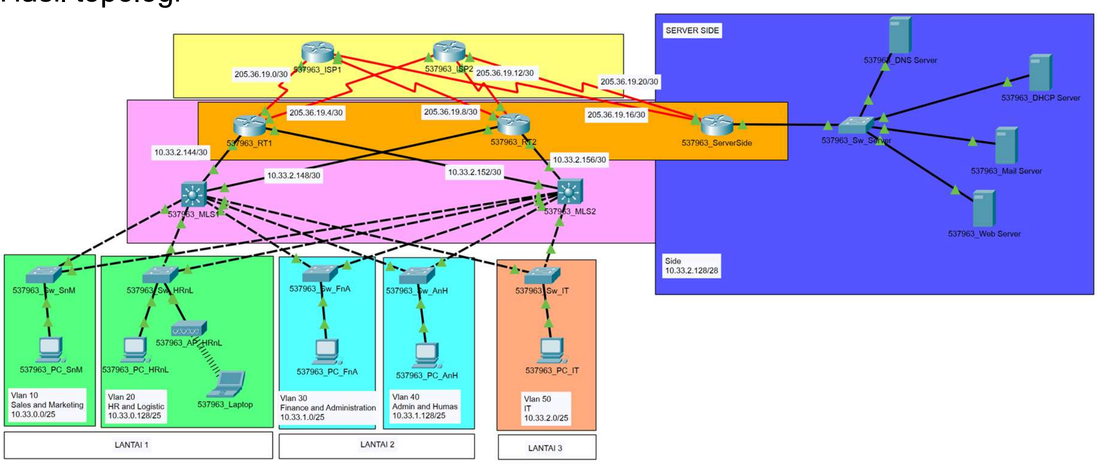


Pool Name	Default Gateway	DNS Server	Start IP Address	Subnet Mask	Max User	TFTP Server	WLC Address
Server	10.33.2.129	10.33.2.131	10.33.2.131	255.255.255.240	10	0.0.0.0	0.0.0.0
IT	10.33.2.1	10.33.2.131	10.33.2.3	255.255.255.128	120	0.0.0.0	0.0.0.0
HR & Logistic	10.33.0.129	10.33.2.121	10.33.0.131	255.255.255.128	120	0.0.0.0	0.0.0.0
Admin & Humas	10.33.1.129	10.33.2.131	10.33.1.131	255.255.255.128	120	0.0.0.0	0.0.0.0
Finansial & Accounting	10.33.1.1	10.33.2.131	10.33.1.3	255.255.255.128	120	0.0.0.0	0.0.0.0
Sales and Marketing	10.33.0.1	10.33.2.131	10.33.0.3	255.255.255.128	120	0.0.0.0	0.0.0.0
serverPool	0.0.0.0	0.0.0.0	10.33.2.128	255.255.255.240	512	0.0.0.0	0.0.0.0

Konfigurasi ini dilakukan melalui GUI untuk menentukan range IP address yang akan diberikan kepada klien secara otomatis, serta menentukan address gateway dan DNS yang akan digunakan oleh klien tersebut.

HASIL PRAKTIKUM

1. Hasil topologi



2. Verifikasi routing
   Router 1

537963_RT1

Physical | Config | CLI | Attributes

IOS Command Line Interface

```
Password:

537963_RT1>en
Password:
537963_RT1#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

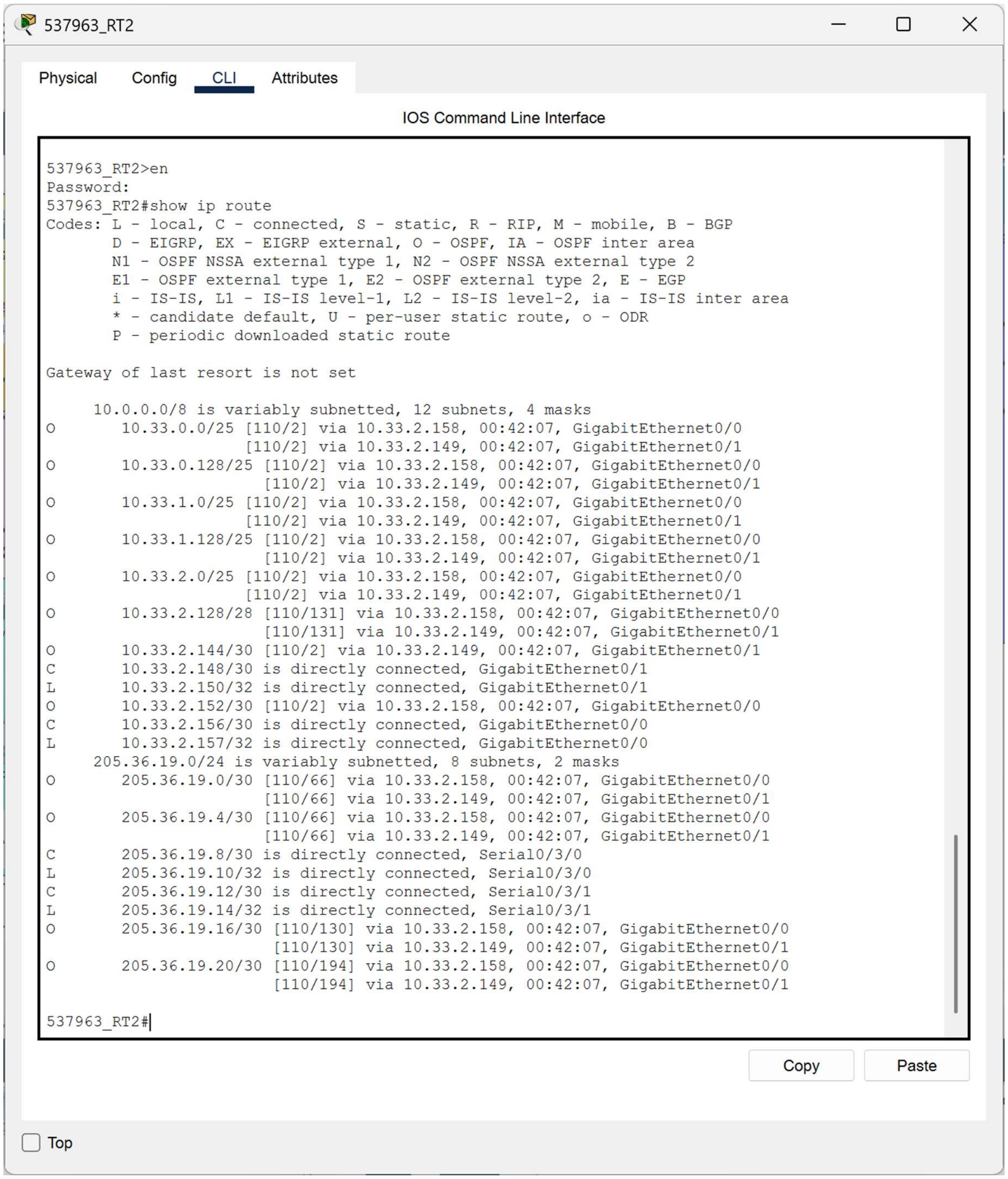
     10.0.0.0/8 is variably subnetted, 12 subnets, 4 masks
O       10.33.0.0/25 [110/2] via 10.33.2.146, 00:41:34, GigabitEthernet0/0
                     [110/2] via 10.33.2.154, 00:41:34, GigabitEthernet0/1
O       10.33.0.128/25 [110/2] via 10.33.2.146, 00:41:34, GigabitEthernet0/0
                       [110/2] via 10.33.2.154, 00:41:34, GigabitEthernet0/1
O       10.33.1.0/25 [110/2] via 10.33.2.146, 00:41:34, GigabitEthernet0/0
                     [110/2] via 10.33.2.154, 00:41:34, GigabitEthernet0/1
O       10.33.1.128/25 [110/2] via 10.33.2.146, 00:41:34, GigabitEthernet0/0
                       [110/2] via 10.33.2.154, 00:41:34, GigabitEthernet0/1
O       10.33.2.0/25 [110/2] via 10.33.2.146, 00:41:34, GigabitEthernet0/0
                     [110/2] via 10.33.2.154, 00:41:34, GigabitEthernet0/1
O       10.33.2.128/28 [110/129] via 205.36.19.1, 00:42:09, Serial0/3/0
C       10.33.2.144/30 is directly connected, GigabitEthernet0/0
L       10.33.2.145/32 is directly connected, GigabitEthernet0/0
O       10.33.2.148/30 [110/2] via 10.33.2.146, 00:41:34, GigabitEthernet0/0
C       10.33.2.152/30 is directly connected, GigabitEthernet0/1
L       10.33.2.153/32 is directly connected, GigabitEthernet0/1
O       10.33.2.156/30 [110/2] via 10.33.2.154, 00:41:34, GigabitEthernet0/1
     205.36.19.0/24 is variably subnetted, 8 subnets, 2 masks
C       205.36.19.0/30 is directly connected, Serial0/3/0
L       205.36.19.2/32 is directly connected, Serial0/3/0
C       205.36.19.4/30 is directly connected, Serial0/3/1
L       205.36.19.6/32 is directly connected, Serial0/3/1
O       205.36.19.8/30 [110/66] via 10.33.2.146, 00:41:34, GigabitEthernet0/0
                       [110/66] via 10.33.2.154, 00:41:34, GigabitEthernet0/1
O       205.36.19.12/30 [110/66] via 10.33.2.146, 00:41:34, GigabitEthernet0/0
                        [110/66] via 10.33.2.154, 00:41:34, GigabitEthernet0/1
O       205.36.19.16/30 [110/128] via 205.36.19.1, 00:42:19, Serial0/3/0
O       205.36.19.20/30 [110/192] via 205.36.19.1, 00:42:09, Serial0/3/0

537963_RT1#
```

Copy | Paste

Top

Router 2


537963_RT2
Physical
Config
CLI
Attributes
IOS Command Line Interface
```
537963_RT2>en
Password:
537963_RT2#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

     10.0.0.0/8 is variably subnetted, 12 subnets, 4 masks
O       10.33.0.0/25 [110/2] via 10.33.2.158, 00:42:07, GigabitEthernet0/0
                     [110/2] via 10.33.2.149, 00:42:07, GigabitEthernet0/1
O       10.33.0.128/25 [110/2] via 10.33.2.158, 00:42:07, GigabitEthernet0/0
                       [110/2] via 10.33.2.149, 00:42:07, GigabitEthernet0/1
O       10.33.1.0/25 [110/2] via 10.33.2.158, 00:42:07, GigabitEthernet0/0
                     [110/2] via 10.33.2.149, 00:42:07, GigabitEthernet0/1
O       10.33.1.128/25 [110/2] via 10.33.2.158, 00:42:07, GigabitEthernet0/0
                       [110/2] via 10.33.2.149, 00:42:07, GigabitEthernet0/1
O       10.33.2.0/25 [110/2] via 10.33.2.158, 00:42:07, GigabitEthernet0/0
                     [110/2] via 10.33.2.149, 00:42:07, GigabitEthernet0/1
O       10.33.2.128/28 [110/131] via 10.33.2.158, 00:42:07, GigabitEthernet0/0
                       [110/131] via 10.33.2.149, 00:42:07, GigabitEthernet0/1
O       10.33.2.144/30 [110/2] via 10.33.2.149, 00:42:07, GigabitEthernet0/1
C       10.33.2.148/30 is directly connected, GigabitEthernet0/1
L       10.33.2.150/32 is directly connected, GigabitEthernet0/1
O       10.33.2.152/30 [110/2] via 10.33.2.158, 00:42:07, GigabitEthernet0/0
C       10.33.2.156/30 is directly connected, GigabitEthernet0/0
L       10.33.2.157/32 is directly connected, GigabitEthernet0/0
     205.36.19.0/24 is variably subnetted, 8 subnets, 2 masks
O       205.36.19.0/30 [110/66] via 10.33.2.158, 00:42:07, GigabitEthernet0/0
                       [110/66] via 10.33.2.149, 00:42:07, GigabitEthernet0/1
O       205.36.19.4/30 [110/66] via 10.33.2.158, 00:42:07, GigabitEthernet0/0
                       [110/66] via 10.33.2.149, 00:42:07, GigabitEthernet0/1
C       205.36.19.8/30 is directly connected, Serial0/3/0
L       205.36.19.10/32 is directly connected, Serial0/3/0
C       205.36.19.12/30 is directly connected, Serial0/3/1
L       205.36.19.14/32 is directly connected, Serial0/3/1
O       205.36.19.16/30 [110/130] via 10.33.2.158, 00:42:07, GigabitEthernet0/0
                        [110/130] via 10.33.2.149, 00:42:07, GigabitEthernet0/1
O       205.36.19.20/30 [110/194] via 10.33.2.158, 00:42:07, GigabitEthernet0/0
                        [110/194] via 10.33.2.149, 00:42:07, GigabitEthernet0/1

537963_RT2#
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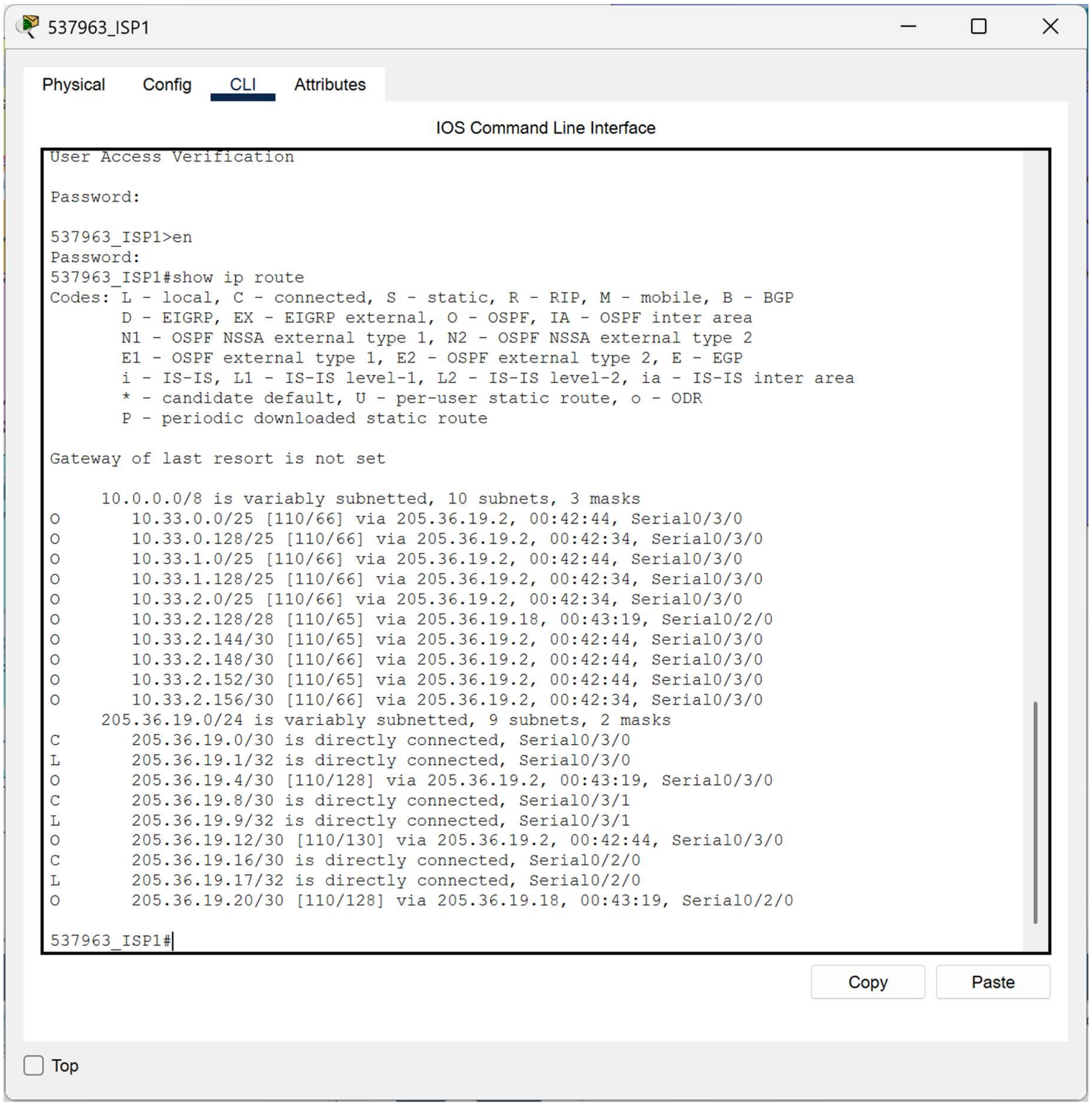
ISP 1


537963_ISP1
Physical
Config
CLI
Attributes
IOS Command Line Interface
User Access Verification

Password:

537963_ISP1>en
Password:
537963_ISP1#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
* - candidate default, U - per-user static route, o - ODR
P - periodic downloaded static route

Gateway of last resort is not set

10.0.0.0/8 is variably subnetted, 10 subnets, 3 masks
O 10.33.0.0/25 [110/66] via 205.36.19.2, 00:42:44, Serial0/3/0
O 10.33.0.128/25 [110/66] via 205.36.19.2, 00:42:34, Serial0/3/0
O 10.33.1.0/25 [110/66] via 205.36.19.2, 00:42:44, Serial0/3/0
O 10.33.1.128/25 [110/66] via 205.36.19.2, 00:42:34, Serial0/3/0
O 10.33.2.0/25 [110/66] via 205.36.19.2, 00:42:34, Serial0/3/0
O 10.33.2.128/28 [110/65] via 205.36.19.18, 00:43:19, Serial0/2/0
O 10.33.2.144/30 [110/65] via 205.36.19.2, 00:42:44, Serial0/3/0
O 10.33.2.148/30 [110/66] via 205.36.19.2, 00:42:44, Serial0/3/0
O 10.33.2.152/30 [110/65] via 205.36.19.2, 00:42:44, Serial0/3/0
O 10.33.2.156/30 [110/66] via 205.36.19.2, 00:42:34, Serial0/3/0
205.36.19.0/24 is variably subnetted, 9 subnets, 2 masks
C 205.36.19.0/30 is directly connected, Serial0/3/0
L 205.36.19.1/32 is directly connected, Serial0/3/0
O 205.36.19.4/30 [110/128] via 205.36.19.2, 00:43:19, Serial0/3/0
C 205.36.19.8/30 is directly connected, Serial0/3/1
L 205.36.19.9/32 is directly connected, Serial0/3/1
O 205.36.19.12/30 [110/130] via 205.36.19.2, 00:42:44, Serial0/3/0
C 205.36.19.16/30 is directly connected, Serial0/2/0
L 205.36.19.17/32 is directly connected, Serial0/2/0
O 205.36.19.20/30 [110/128] via 205.36.19.18, 00:43:19, Serial0/2/0

537963_ISP1#
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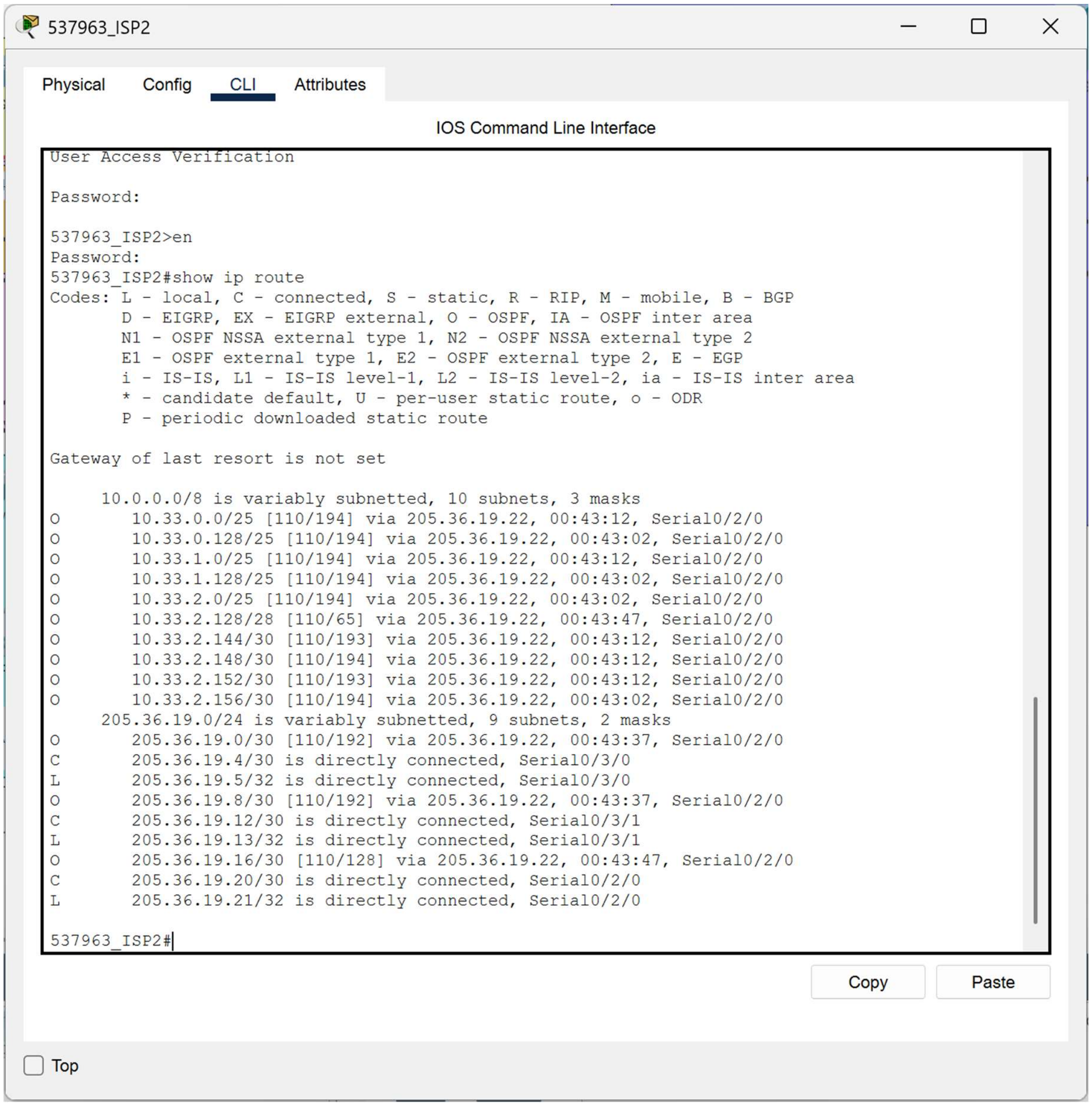
ISP 2


537963_ISP2
Physical
Config
CLI
Attributes
IOS Command Line Interface
User Access Verification
Password:
537963_ISP2>en
Password:
537963_ISP2#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
* - candidate default, U - per-user static route, o - ODR
P - periodic downloaded static route
Gateway of last resort is not set
10.0.0.0/8 is variably subnetted, 10 subnets, 3 masks
O 10.33.0.0/25 [110/194] via 205.36.19.22, 00:43:12, Serial0/2/0
O 10.33.0.128/25 [110/194] via 205.36.19.22, 00:43:02, Serial0/2/0
O 10.33.1.0/25 [110/194] via 205.36.19.22, 00:43:12, Serial0/2/0
O 10.33.1.128/25 [110/194] via 205.36.19.22, 00:43:02, Serial0/2/0
O 10.33.2.0/25 [110/194] via 205.36.19.22, 00:43:02, Serial0/2/0
O 10.33.2.128/28 [110/65] via 205.36.19.22, 00:43:47, Serial0/2/0
O 10.33.2.144/30 [110/193] via 205.36.19.22, 00:43:12, Serial0/2/0
O 10.33.2.148/30 [110/194] via 205.36.19.22, 00:43:12, Serial0/2/0
O 10.33.2.152/30 [110/193] via 205.36.19.22, 00:43:12, Serial0/2/0
O 10.33.2.156/30 [110/194] via 205.36.19.22, 00:43:02, Serial0/2/0
205.36.19.0/24 is variably subnetted, 9 subnets, 2 masks
O 205.36.19.0/30 [110/192] via 205.36.19.22, 00:43:37, Serial0/2/0
C 205.36.19.4/30 is directly connected, Serial0/3/0
L 205.36.19.5/32 is directly connected, Serial0/3/0
O 205.36.19.8/30 [110/192] via 205.36.19.22, 00:43:37, Serial0/2/0
C 205.36.19.12/30 is directly connected, Serial0/3/1
L 205.36.19.13/32 is directly connected, Serial0/3/1
O 205.36.19.16/30 [110/128] via 205.36.19.22, 00:43:47, Serial0/2/0
C 205.36.19.20/30 is directly connected, Serial0/2/0
L 205.36.19.21/32 is directly connected, Serial0/2/0
537963_ISP2#
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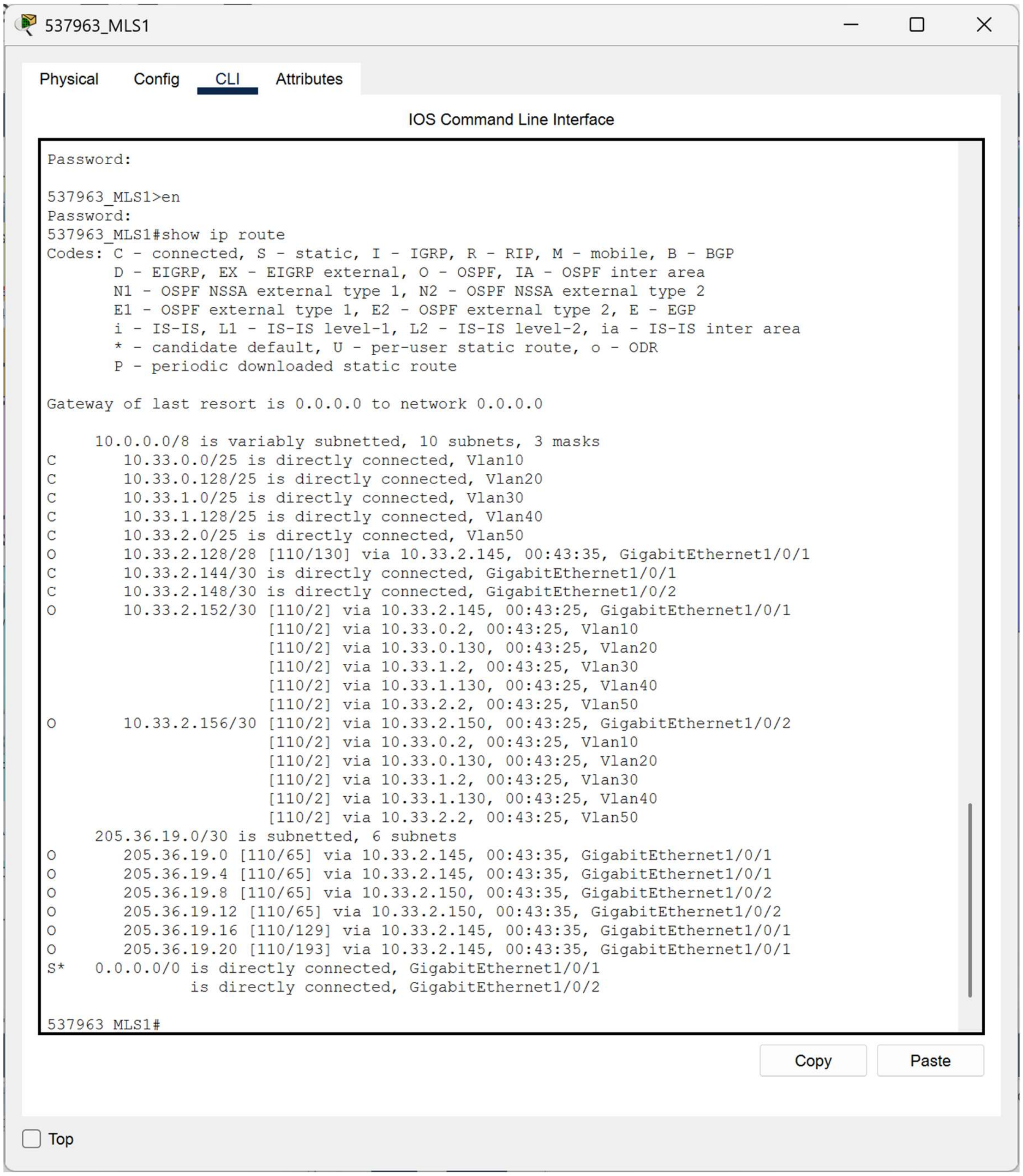
Paste
Top

MLS 1




```
Password:

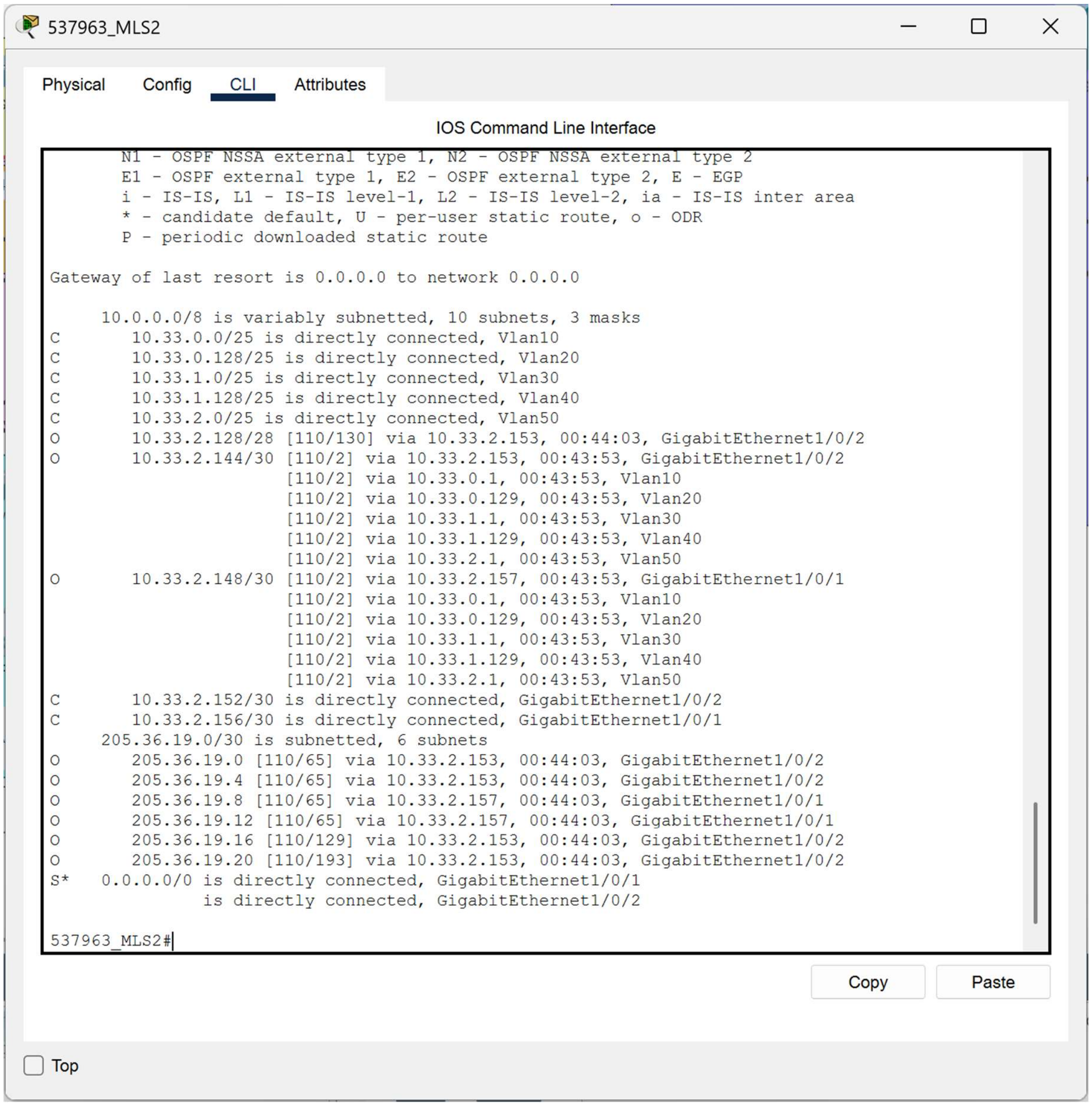
537963_MLS1>en
Password:
537963_MLS1#show ip route
Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is 0.0.0.0 to network 0.0.0.0

     10.0.0.0/8 is variably subnetted, 10 subnets, 3 masks
C       10.33.0.0/25 is directly connected, Vlan10
C       10.33.0.128/25 is directly connected, Vlan20
C       10.33.1.0/25 is directly connected, Vlan30
C       10.33.1.128/25 is directly connected, Vlan40
C       10.33.2.0/25 is directly connected, Vlan50
O       10.33.2.128/28 [110/130] via 10.33.2.145, 00:43:35, GigabitEthernet1/0/1
C       10.33.2.144/30 is directly connected, GigabitEthernet1/0/1
C       10.33.2.148/30 is directly connected, GigabitEthernet1/0/2
O       10.33.2.152/30 [110/2] via 10.33.2.145, 00:43:25, GigabitEthernet1/0/1
                       [110/2] via 10.33.0.2, 00:43:25, Vlan10
                       [110/2] via 10.33.0.130, 00:43:25, Vlan20
                       [110/2] via 10.33.1.2, 00:43:25, Vlan30
                       [110/2] via 10.33.1.130, 00:43:25, Vlan40
                       [110/2] via 10.33.2.2, 00:43:25, Vlan50
O       10.33.2.156/30 [110/2] via 10.33.2.150, 00:43:25, GigabitEthernet1/0/2
                       [110/2] via 10.33.0.2, 00:43:25, Vlan10
                       [110/2] via 10.33.0.130, 00:43:25, Vlan20
                       [110/2] via 10.33.1.2, 00:43:25, Vlan30
                       [110/2] via 10.33.1.130, 00:43:25, Vlan40
                       [110/2] via 10.33.2.2, 00:43:25, Vlan50
     205.36.19.0/30 is subnetted, 6 subnets
O       205.36.19.0 [110/65] via 10.33.2.145, 00:43:35, GigabitEthernet1/0/1
O       205.36.19.4 [110/65] via 10.33.2.145, 00:43:35, GigabitEthernet1/0/1
O       205.36.19.8 [110/65] via 10.33.2.150, 00:43:35, GigabitEthernet1/0/2
O       205.36.19.12 [110/65] via 10.33.2.150, 00:43:35, GigabitEthernet1/0/2
O       205.36.19.16 [110/129] via 10.33.2.145, 00:43:35, GigabitEthernet1/0/1
O       205.36.19.20 [110/193] via 10.33.2.145, 00:43:35, GigabitEthernet1/0/1
S*   0.0.0.0/0 is directly connected, GigabitEthernet1/0/1
               is directly connected, GigabitEthernet1/0/2

537963_MLS1#
```

MLS 2


537963_MLS2
Physical
Config
CLI
Attributes
IOS Command Line Interface
N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
* - candidate default, U - per-user static route, o - ODR
P - periodic downloaded static route
Gateway of last resort is 0.0.0.0 to network 0.0.0.0
10.0.0.0/8 is variably subnetted, 10 subnets, 3 masks
C 10.33.0.0/25 is directly connected, Vlan10
C 10.33.0.128/25 is directly connected, Vlan20
C 10.33.1.0/25 is directly connected, Vlan30
C 10.33.1.128/25 is directly connected, Vlan40
C 10.33.2.0/25 is directly connected, Vlan50
O 10.33.2.128/28 [110/130] via 10.33.2.153, 00:44:03, GigabitEthernet1/0/2
O 10.33.2.144/30 [110/2] via 10.33.2.153, 00:43:53, GigabitEthernet1/0/2
[110/2] via 10.33.0.1, 00:43:53, Vlan10
[110/2] via 10.33.0.129, 00:43:53, Vlan20
[110/2] via 10.33.1.1, 00:43:53, Vlan30
[110/2] via 10.33.1.129, 00:43:53, Vlan40
[110/2] via 10.33.2.1, 00:43:53, Vlan50
O 10.33.2.148/30 [110/2] via 10.33.2.157, 00:43:53, GigabitEthernet1/0/1
[110/2] via 10.33.0.1, 00:43:53, Vlan10
[110/2] via 10.33.0.129, 00:43:53, Vlan20
[110/2] via 10.33.1.1, 00:43:53, Vlan30
[110/2] via 10.33.1.129, 00:43:53, Vlan40
[110/2] via 10.33.2.1, 00:43:53, Vlan50
C 10.33.2.152/30 is directly connected, GigabitEthernet1/0/2
C 10.33.2.156/30 is directly connected, GigabitEthernet1/0/1
205.36.19.0/30 is subnetted, 6 subnets
O 205.36.19.0 [110/65] via 10.33.2.153, 00:44:03, GigabitEthernet1/0/2
O 205.36.19.4 [110/65] via 10.33.2.153, 00:44:03, GigabitEthernet1/0/2
O 205.36.19.8 [110/65] via 10.33.2.157, 00:44:03, GigabitEthernet1/0/1
O 205.36.19.12 [110/65] via 10.33.2.157, 00:44:03, GigabitEthernet1/0/1
O 205.36.19.16 [110/129] via 10.33.2.153, 00:44:03, GigabitEthernet1/0/2
O 205.36.19.20 [110/193] via 10.33.2.153, 00:44:03, GigabitEthernet1/0/2
S* 0.0.0.0/0 is directly connected, GigabitEthernet1/0/1
is directly connected, GigabitEthernet1/0/2
537963_MLS2#
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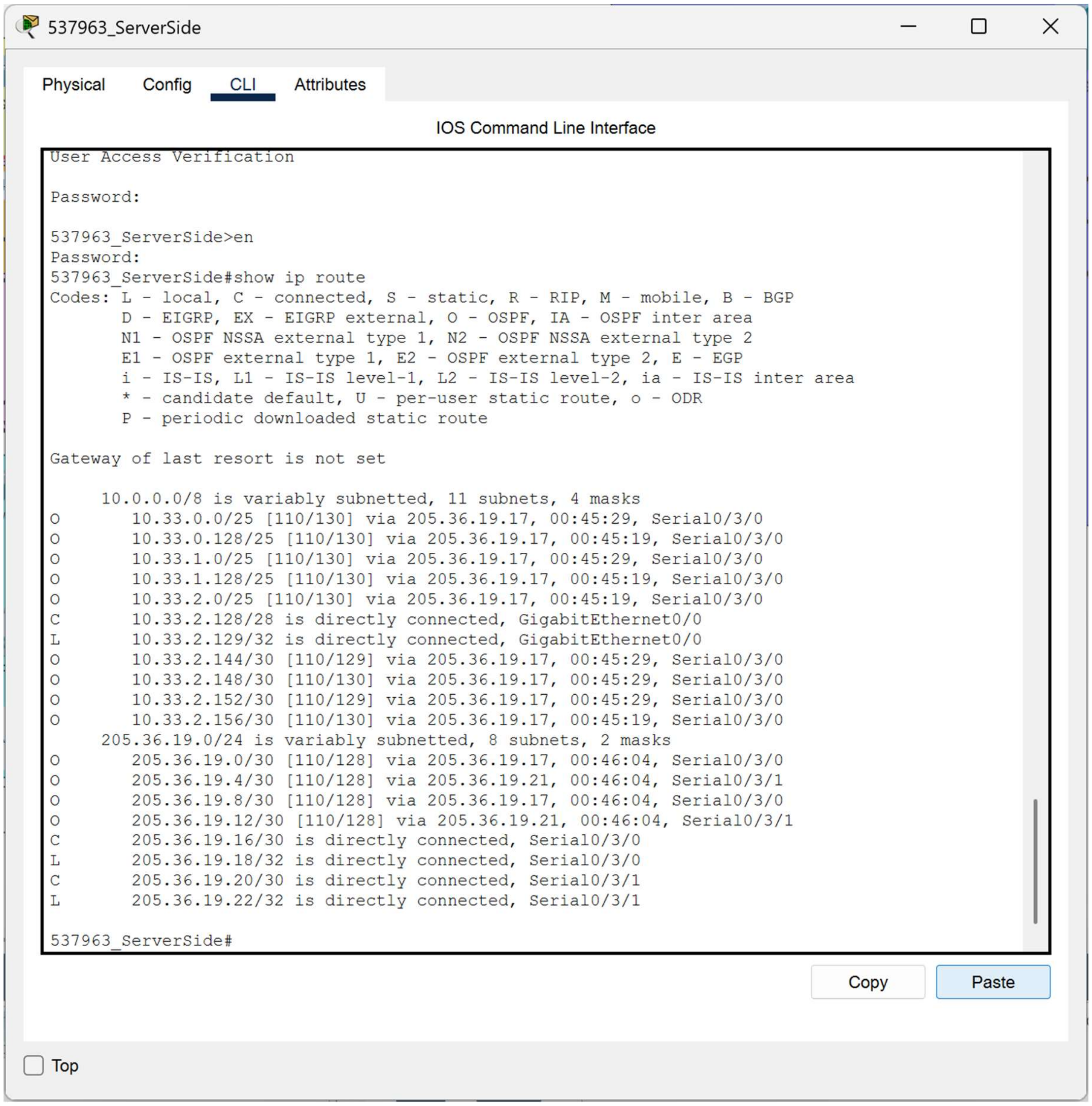
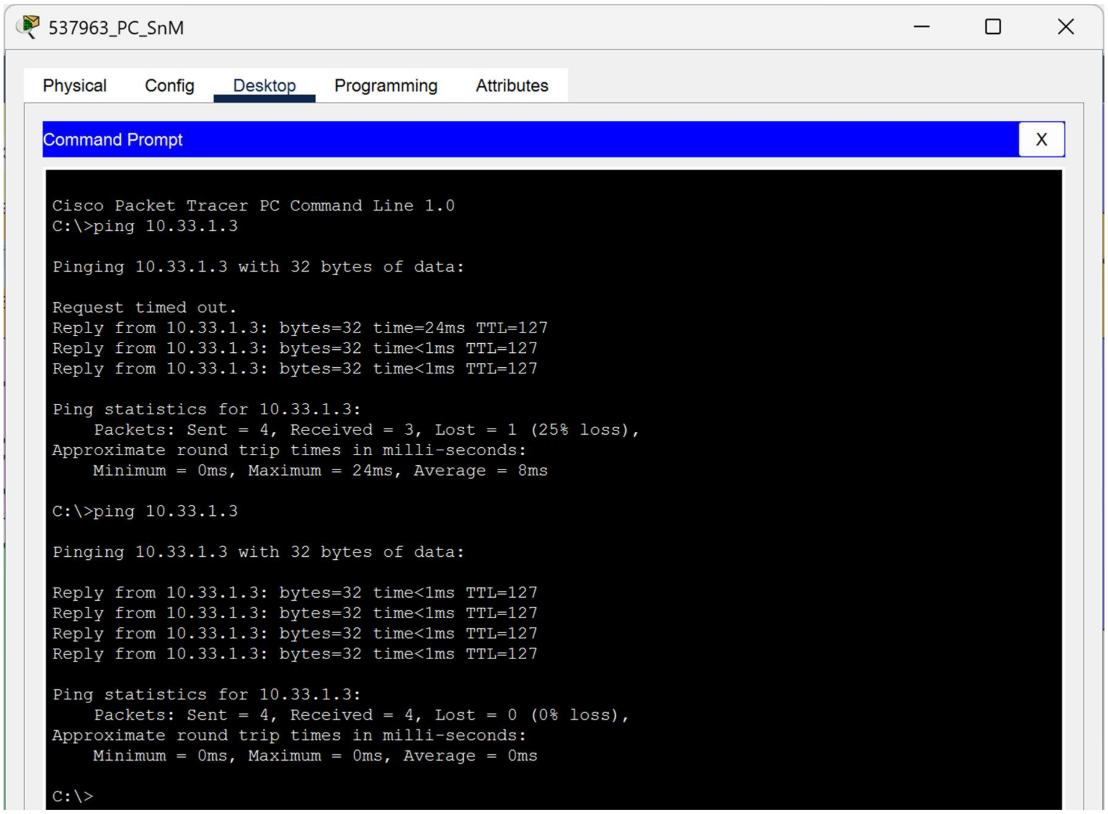
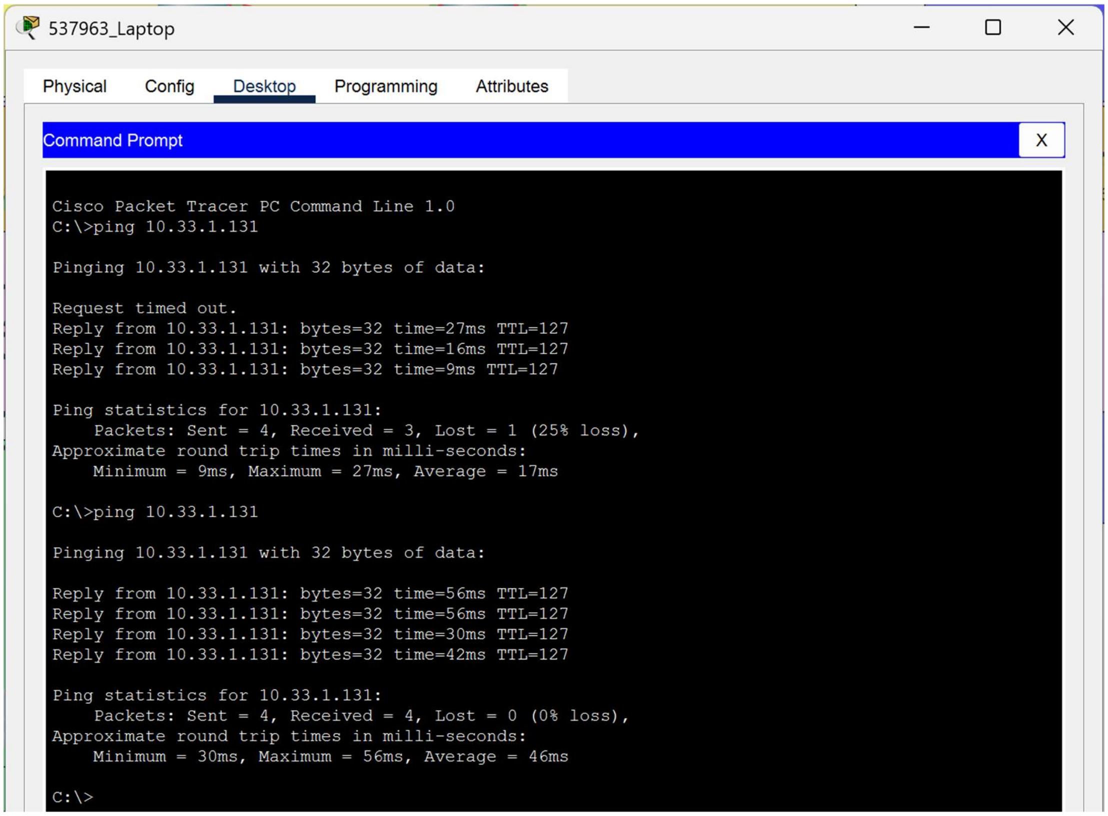
Server Side



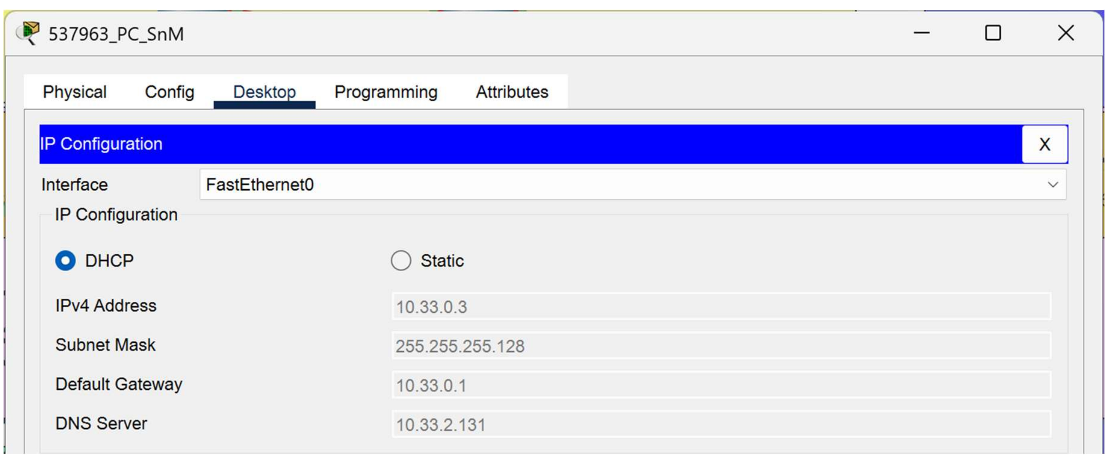
3. Uji koneksi - Ping dari salah satu komputer ke komputer lain **(jelaskan dari perangkat mana ke perangkat mana)**
   PC SnM ke PC FnA



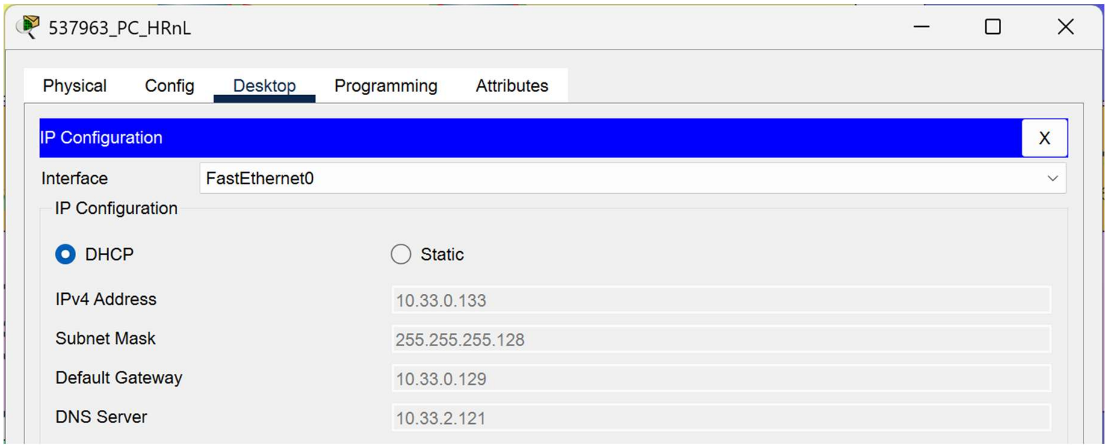
4. Uji koneksi wireless - Ping dari perangkat laptop ke salah satu perangkat lain.
   Laptop ke PC AnH



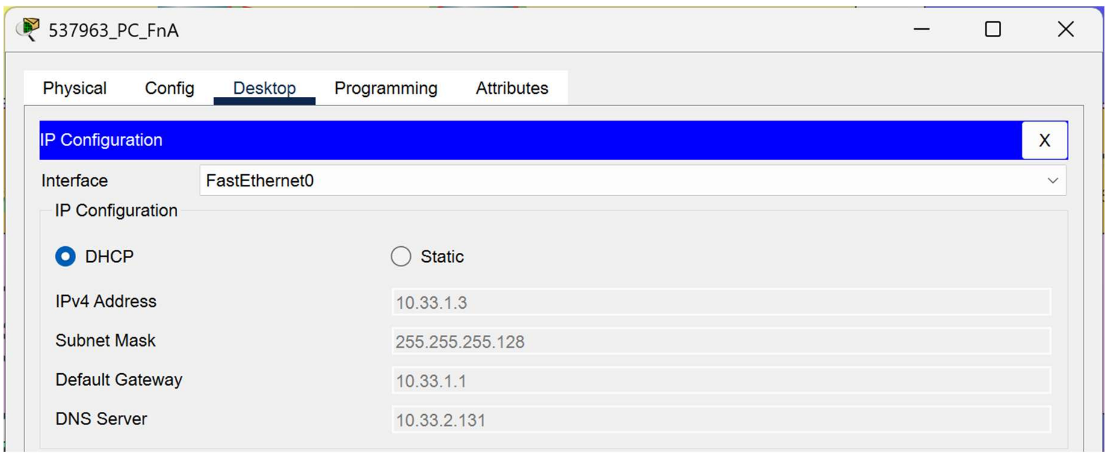
5. Verifikasi DHCP
   PC SnM



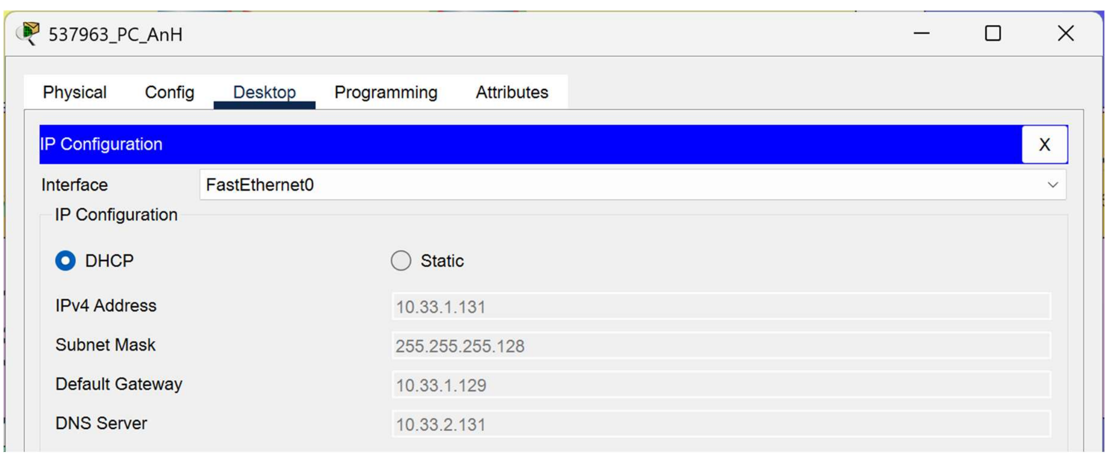
PC HRnL



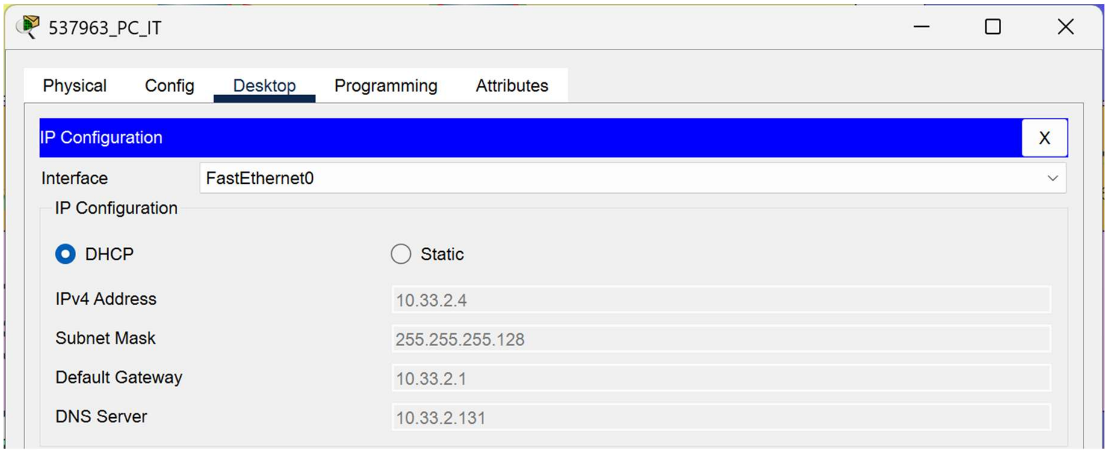
PC FnA



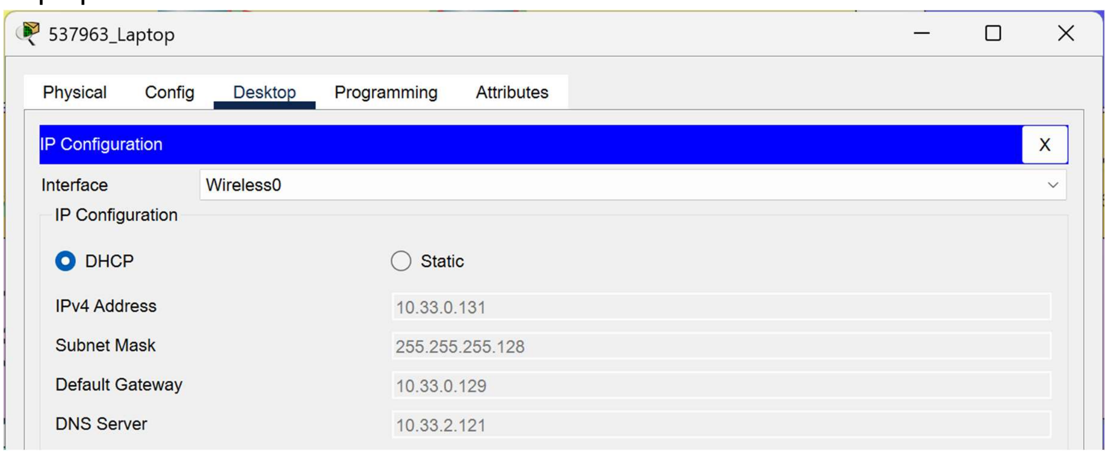
PC AnH



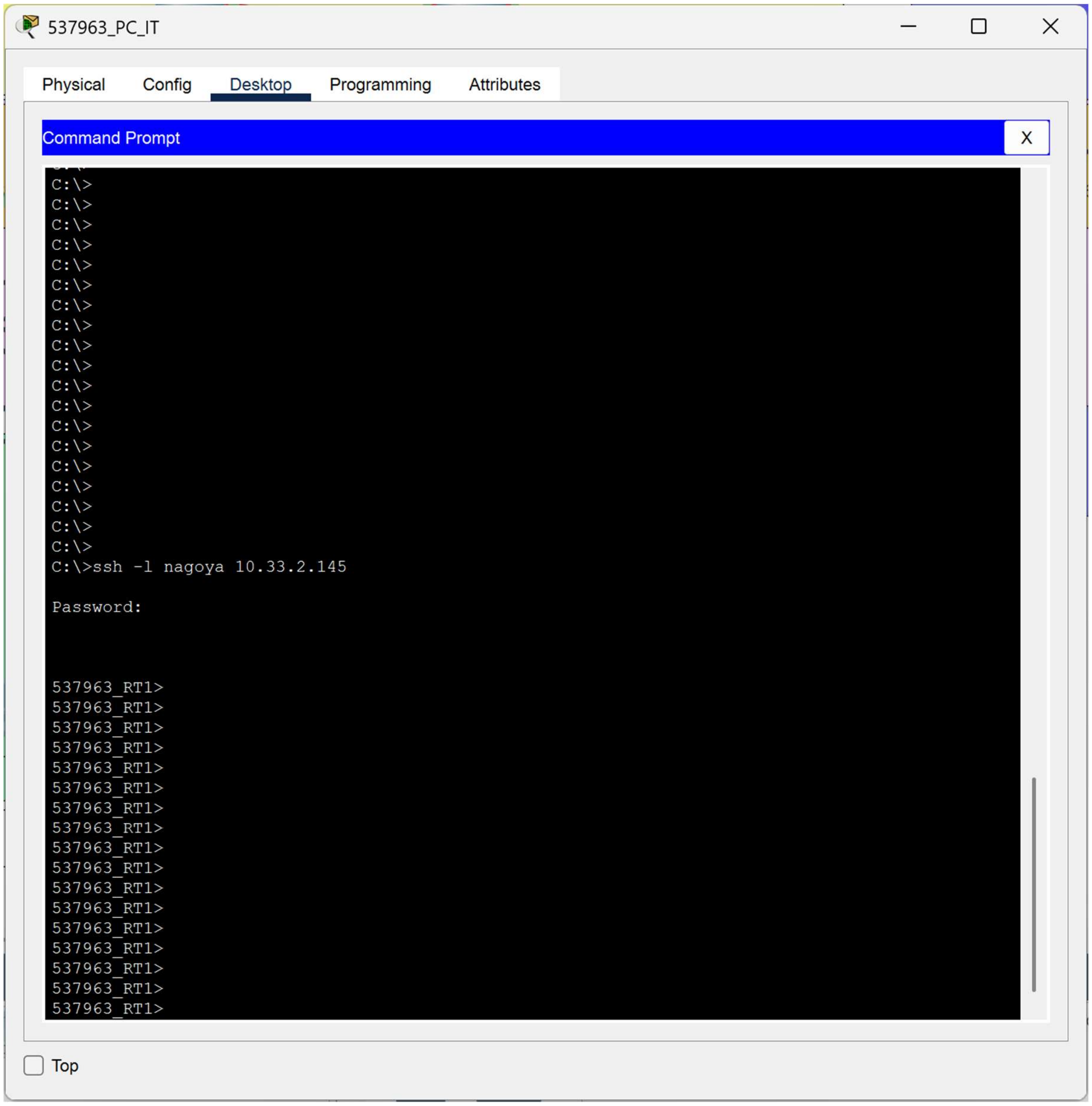
PC IT



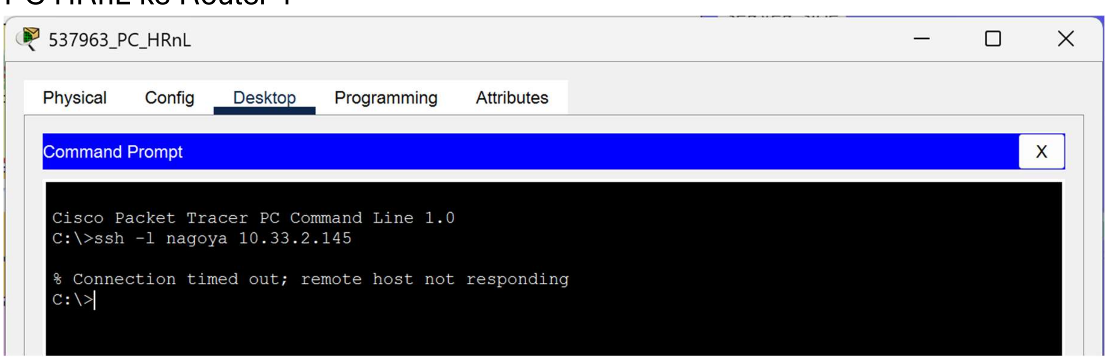
Laptop



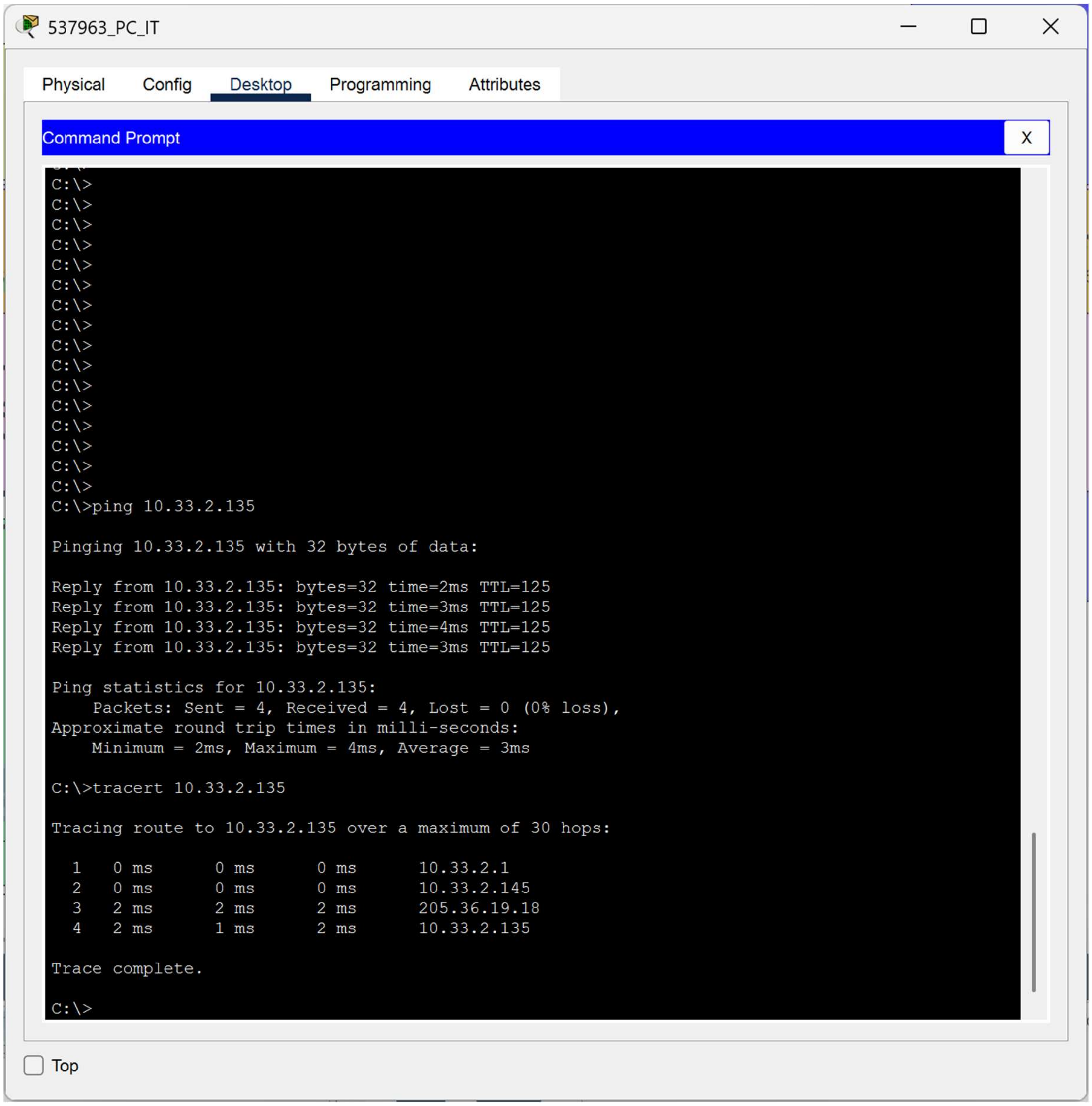
6. Uji SSH-ACL dari salah satu komputer ke salah satu perangkat router
   PC IT ke Router 1



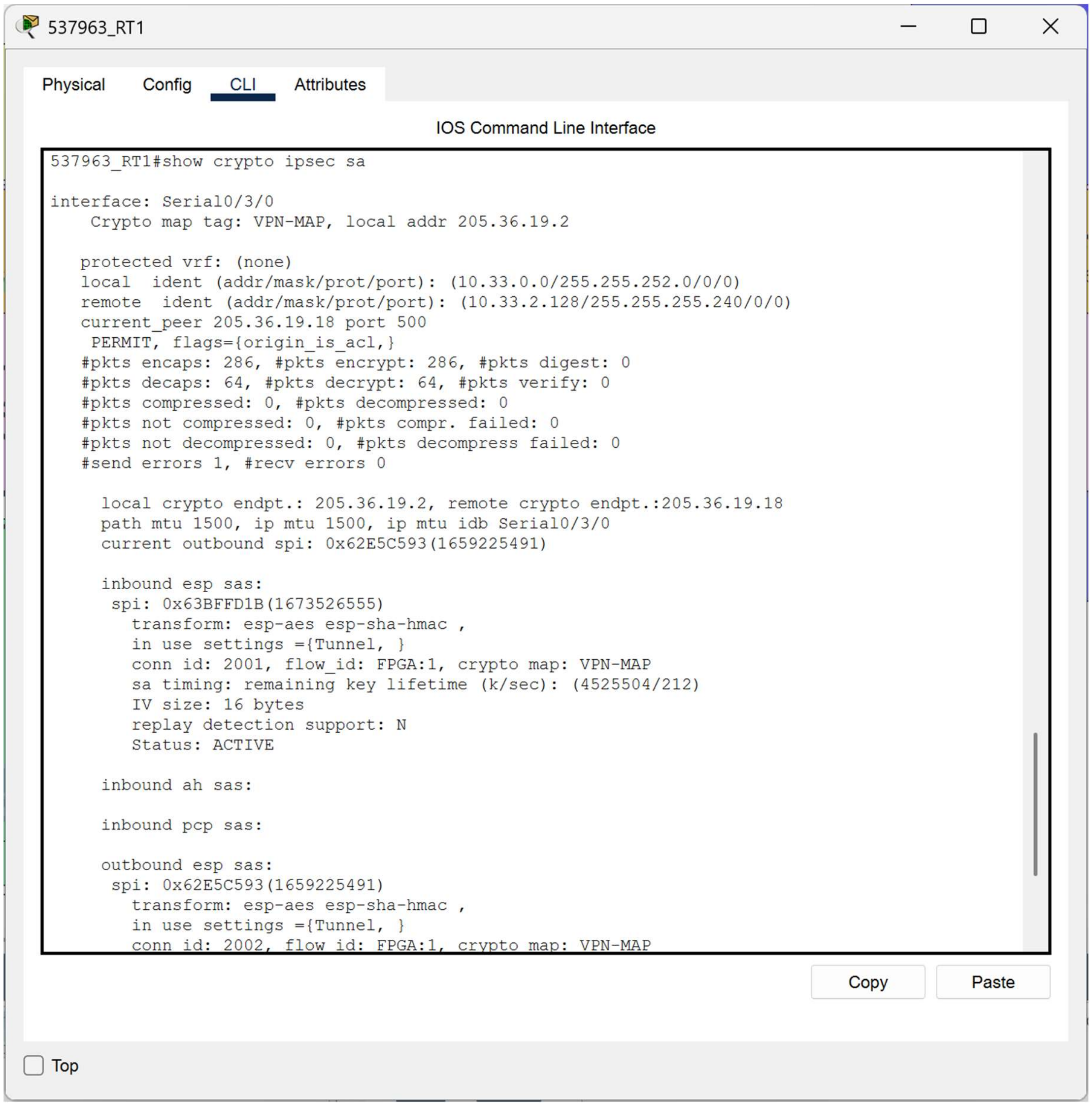
PC HRnL ke Router 1



7. Uji IPSec-VPN
   Koneksi PC IT ke DHCP Server



IPsec Router 1



PEMBAHASAN

1. Keamanan akses perangkat

   Pada tahap awal, pengamanan remote management pada 537963_RT1 menjadi prioritas untuk mencegah akses tidak sah. Konfigurasi dilakukan dengan mengaktifkan SSH v2 yang memerlukan domain-name dan crypto key RSA sebesar 1024 bit. Untuk memperketat akses, diterapkan access-list 10 pada line vty 0 15. Berdasarkan hasil pengujian di lapangan, PC IT yang berada dalam rentang IP yang diizinkan berhasil masuk ke sistem, sementara PC HRnL mengalami connection timed out. Hal ini membuktikan bahwa kebijakan ACL telah berhasil mem-filter trafik manajemen hanya dari segmen yang terpercaya.
2. Implementasi IPSec VPN

Koneksi antara kantor dan data center (server side) dirancang menggunakan tunnel IPSec VPN untuk menjamin integritas data saat melewati jaringan publik (ISP). Langkah wajib yang dilakukan sebelum konfigurasi adalah mengaktifkan lisensi securityk9 pada Router 1 dan melakukan reload perangkat agar fitur kriptografi tersedia.

Konfigurasi VPN dibagi menjadi dua fase: fase pertama (ISAKMP) untuk negosiasi kunci keamanan dan fase kedua (IPSec) untuk enkripsi data aktual.

3. Routing OSPF

   Protokol OSPF digunakan untuk memastikan konvergensi rute secara dinamis di seluruh jaringan. Verifikasi jalur dilakukan dengan perintah tracert dari PC IT ke DHCP Server (10.33.2.135). Hasilnya menunjukkan bahwa paket melompat dari gateway lokal (10.33.2.1) langsung menuju IP publik Router ServerSide (205.36.19.18) tanpa melewati ISP secara detail. Ini membuktikan bahwa mekanisme tunneling dan rute statis yang mengarah ke gateway ISP sudah berjalan secara sinkron

4. Troubleshooting DHCP

   Salah satu poin krusial dalam praktikum ini adalah penyelesaian masalah distribusi IP otomatis (DHCP). Awalnya, klien gagal mendapatkan IP meskipun rute VPN sudah up. Melalui proses troubleshooting mendalam pada sisi Data Center, ditemukan bahwa masalah utama bukan pada routing, melainkan pada konfigurasi switch bagian server side.

   Kendala tersebut diatasi dengan mengatur ulang mode interface. Port fa0/1-2 yang mengarah ke router diatur menjadi mode Trunk agar dapat melewatkan trafik dari berbagai VLAN, sementara port fa0/3-24 dikunci pada mode Access untuk end devices (server). Setelah konfigurasi ini dilakukan, layanan DHCP langsung berjalan normal, dibuktikan dengan PC SnM, PC HRnL, dan PC AnH yang berhasil mendapatkan konfigurasi IP, Gateway, serta DNS Server secara otomatis dari server pusat.

DAFTAR PUSTAKA

Oppenheimer, Priscilla. (2011). *Top-Down Network Design: A Systems Analysis Approach to Enterprise Network Design*. Edisi ke-3. Indianapolis: Cisco Press

Tim Dosen Departemen Teknik Elektro dan Informatika. (2026). *Modul Praktikum Perancangan Jaringan: Pemilihan dan Penerapan Protokol Switching dan Routing*. Yogyakarta: Sekolah Vokasi Universitas Gadjah Mada